



SIMULATION OF MATERNAL INSTINCT

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Simulation of Maternal Instinct

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Abstract

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Maternal care is a foundational form of altruism, yet the mechanisms by which prosocial strategies emerge and persist in artificial systems remain unclear. This thesis presents a computational framework that couples a neuroendocrine-inspired controller with evolutionary agent-based modeling to study how caregiving-like behavior can emerge in a minimal dyadic setting. Building on evidence for oxytocin- and cortisol-like modulation of social bonding and stress responses, we model a mother agent whose perception $\rightarrow$ hormone $\rightarrow$ motivation cascade produces behavior through weighted motivational competition (Forage, Care, Self, Protect). The child agent is passive: it exposes internal need variables (hunger, warmth, injury) that the mother observes and acts on, creating an asymmetric caregiving dynamic without explicit cooperative rewards.

We run spatially explicit simulations in a 15×15 grid-world with periodic food spawning (food spawn interval $\in \{5, 15, 25\}$ ticks; shorter interval = more abundant food) and optional threats. Motivation-weight genomes are optimized using a single-lineage (1+1) evolutionary search evaluated over 10 stochastic rollouts per candidate, across 3000 generations. We compare three plasticity regimes—E2 (genes only), P1 (adaptive-rate outcome plasticity), and P2 (fixed-rate outcome plasticity)—and three motivation-weight initialization priors: `anti_maternal`, `random_uniform`, and `pro_maternal`.

Four experiments are reported: (1) characterizing maternal behavior dynamics in the training ecology; (2) showing that plasticity accelerates evolutionary fitness improvement (Baldwin-compatible pattern); (3) demonstrating that initialization priors shape weight displacement and residual failure modes after 3000 generations; and (4) evaluating generalization to nine unseen world $\times$ food cells, showing graded transfer: survival is maintained under abundant food and low threat, but child hunger becomes the dominant failure mode under scarce food (interval 25) and high threat.

The framework bridges descriptive biology and abstract cooperation models, offering mechanistic insight into how minimal neuroendocrine architectures generate adaptive prosocial strategies.

Keywords: maternal behavior, neuroendocrine modulation, evolutionary computation, agent-based modeling, motivation weights, Baldwin effect, bio-inspired AI

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List of Symbols

Symbol	Description	Unit / Type
$\mathbf{u}$	Motivation-weight genome vector of the mother agent	–
u_k	Individual motivation weight for action class k	[0,1]
g	Generation index in the evolutionary search	–
ϵ	Gaussian mutation noise; $\epsilon \sim \mathcal{N}(0, 0.05^2)$	–
F	Composite fitness score of a candidate genome	[0,1]
α	Child-fitness weighting coefficient in fitness function ($\alpha = 0.5$)	–
λ	Injury penalty coefficient in fitness function ($\lambda = 0.002$)	–
TTD_m	Normalized time-to-death of the mother	[0,1]
TTD_c	Normalized time-to-death of the child	[0,1]
I_{total}	Cumulative injury accumulated by the mother over an episode	–
L_2	Euclidean (L2) distance between two genome weight vectors	–
N_s	Number of stochastic rollouts per genome evaluation ($N_s = 10$)	–
N_{seed}	Number of independent evolution seeds ($N_{\text{seed}} = 30$)	–
T	Episode length in simulation ticks ($T = 1000$)	ticks
η	Plasticity learning rate (P1: adaptive; P2: fixed)	–
d_{global}	Global deficit signal used by P1 plasticity regime	–
d_k	Local deficit signal for action class k , used by P2	–
OT	Oxytocin-like modulator signal (bonding/affiliation)	a.u.
CORT	Cortisol-like modulator signal (stress/threat response)	a.u.
$P(a \mid s)$	Conditional probability of selecting action a given state s	[0,1]
w_{50}	Window size for rolling average of L2 genome drift (50 generations)	gen

List of Technical Vocabulary and Abbreviations

Term / Abbreviation	Definition
ABM	= Agent-Based Model
AI	= Artificial Intelligence
CORT	= Cortisol-like modulator (stress/threat signal in the model)
E2	= Evolution-only plasticity regime; only the genome is updated, no within-lifetime weight adjustment
Exp1	= Experiment 1: characterization of maternal behavior in training ecology
Exp2	= Experiment 2: comparison of plasticity regimes (Baldwin effect)
Exp3	= Experiment 3: effect of motivation-weight initialization priors
Exp4	= Experiment 4: generalization to unseen food and threat conditions
IQR	= Interquartile Range ($Q3 - Q1$); used as spread measure for skewed distributions
LR	= Learning Rate; controls magnitude of within-lifetime weight updates
OT	= Oxytocin-like modulator (bonding/affiliation signal in the model)
P1	= Plasticity regime 1: adaptive learning rate, global deficit signal
P2	= Plasticity regime 2: fixed learning rate, local (per-action) deficit signal
SEM	= Standard Error of the Mean; $SEM = \sigma / \sqrt{N}$
TTD	= Time-To-Death; the simulation tick at which an agent dies, normalized by episode length
<i>Behavioral actions (mother)</i>	
Care	= Maternal care action: attending to the child's needs (warmth, injury)
Forage	= Foraging action: moving to collect food for the mother or child
Protect	= Protective action: moving to intercept threats approaching the child
Self	= Self-maintenance action: resting or satisfying the mother's own energy needs
<i>Initialization regimes (Exp3)</i>	
anti_maternal	= Initial genome biased away from Care and Protect (low maternal weights)
random_uniform	= Initial genome drawn uniformly at random
pro_maternal	= Initial genome biased toward Care and Protect (high maternal weights)

Chapter 1 Introduction

1.1 Background

“The only way we can safely live with AI in the long term may be to give it something like maternal instinct.”

— GEOFFREY HINTON, quoted in *Fortune*, Aug 2025 [1]

Geoffrey Hinton, often called the ‘Godfather of AI’, suggests that as artificial intelligence grows more powerful, traditional control methods, such as hard-coded rules or external constraints will not be enough to ensure safety. Instead, AI should be endowed with an intrinsic drive seen in maternal behavior. In Hinton’s view, only by embedding empathy and care at a fundamental level can humans coexist safely with advanced AI systems. But what does ‘maternal instinct’ mean in computational terms? And can it be engineered or must it evolve?

Maternal instinct is not a single trait, but an evolved behavioral architecture shaped by millions of years of selection pressure. It encompasses rapid threat detection and protective responses, allocation of limited resources toward vulnerable dependents, tolerance of personal cost for offspring welfare, flexible adaptation to environmental conditions, and critically for social species the capacity to extend caregiving beyond one’s own genetic offspring. This last feature, alloparental care or ‘altruistic nurturing’ forms the biological foundation for broader prosocial behavior in human and animal societies.

For artificial systems, replicating this architecture presents a fundamental challenge. Unlike programmed ethical rules or learned reward functions, biological maternal strategies emerge from the interaction of genetic predispositions, hormonal modulation, neural circuits, and environmental pressures throughout evolutionary time. The result is not rigid altruism but adaptive cooperative agents that balance self-interest with group welfare depending on context.

The importance of maternal care extends beyond biology into culture and symbolism. Across human societies, the archetype of “Mother” represents unconditional care, selfless protection, and the willingness to prioritize the welfare of another over one’s own. From religious iconography to philosophical ideals, this figure embodies humanity’s recognition that civilization itself rests on the capacity to care for the vulnerable. Evolutionary psychologists argue this is no accident the neural and hormonal circuits that evolved for infant care became the cognitive scaffold for broader empathy and moral reasoning. Maternal instinct, in this view, is not merely about raising offspring it is the biological root of all prosocial behavior, the evolutionary foundation upon which human cooperation is built.

Recent advances in neuroscience highlight how maternal behavior emerges from layered biological mechanisms: genetic variation in oxytocin receptors creates baseline prosocial tendencies, hormonal systems dynamically modulate these tendencies based

on stress and reproductive state, and neural circuits integrate infant cues with threat assessment to guide rapid caregiving decisions. This architecture provides a blueprint for artificial agents.

However, simply mimicking these mechanisms in isolation is insufficient. In biological systems, the effectiveness of maternal strategies depends on their fit with environmental conditions. Kin-selective care dominates when resources are abundant and threats minimal, cooperative breeding emerges in harsh environments where group defense provides survival advantages, competitive territoriality arises when resources are scarce and defensible. The context dependence suggests that prosocial behavior cannot be hard-coded but must emerge through selection pressures that favor strategies adaptive to specific ecological niches.

This thesis investigates whether maternal-like caregiving behavior can emerge in artificial agents through biologically inspired internal regulation mechanisms. Rather than explicitly programming cooperative behavior through reward engineering or fixed rules, the proposed framework models maternal decision making as the interaction between homeostatic internal states, hormone-like modulators, and competing motivational drives within an agent-based simulation environment.

In the proposed model, maternal agents maintain a set of internal physiological and psychological states representing simplified biological processes. These include energy and fatigue as homeostatic variables, and hormone-like signals such as oxytocin and cortisol that modulate caregiving and stress responses. These internal states influence a motivational system in which multiple behavioral drives—such as foraging, caregiving, self-recovery, and protection—compete for action selection.

Behavior is determined through a weighted integration of internal signals rather than explicit reward optimization. The agent selects actions by choosing the motivation with the highest activation at each timestep. To allow adaptation over time, the system includes plasticity mechanisms that adjust the influence of internal signals on motivations based on behavioral outcomes. Through repeated interaction with the environment, maternal strategies emerge from the dynamic interaction between internal regulation, environmental conditions, and learning.

In addition to lifetime plasticity, the thesis studies evolutionary adaptation of inherited motivation weights. In this implementation, evolution is performed as a single-lineage search: at each generation a candidate genome is produced by Gaussian mutation, evaluated over multiple stochastic rollouts (different random seeds), and accepted only if mean fitness improves. This captures selection pressure across generations while remaining computationally tractable.

In essence, this research addresses Hinton’s challenge by operationalizing ‘maternal instinct’ as an emergent property of evolutionary dynamics rather than an engineered feature. By grounding agent design in the biological mechanisms that produce adaptive maternal behavior—genetic encoding, hormonal modulation, neural decision-making—and subjecting these agents to selection pressures analogous to those faced by biological mothers, we can identify the minimal requirements for prosocial AI. The question is no longer whether machines can be programmed to care, but whether the capacity for genuine, context-appropriate compassion can arise through the same evolutionary processes that produced it in nature. This thesis provides a computational framework to test that hypothesis, offering insights into how future AI systems might develop the intrinsic motivation

for cooperation that Hinton argues is essential for safe long-term human-AI coexistence.

The resulting architecture integrates three components: internal physiological regulation, a motivational decision system, and evolutionary adaptation across generations. Together these mechanisms form a biologically inspired framework for studying how caregiving behavior can emerge in artificial agents.

Operational definition of “emergence” in this thesis. Because caregiving is not directly rewarded by an external task objective, we treat maternal behavior as *emergent* when it becomes measurable through convergent changes in outcomes and conditional behaviors. Concretely, emergence is assessed by (i) improved normalized time-to-death (TTD) for the child and mother and reduced child injury under the fitness function, (ii) increased conditional action probabilities that align with child needs (e.g., higher $P(\text{Forage} \mid \text{hungry})$, $P(\text{Care} \mid \text{cold})$, $P(\text{Protect} \mid \text{injured})$), and (iii) systematic shifts in evolved motivation weights away from anti-maternal or random initializations. This multi-view definition avoids attributing emergence to a single metric and supports interpretation of *why* survival improves.

1.2 Problem Statement

1.2.1 Current Challenges

As artificial intelligence systems become increasingly autonomous and integrated into human environments, the challenge of creating artificial agents capable of exhibiting genuine prosocial behavior—such as cooperation, empathy, and altruism—has become critical. Current approaches to multi-agent cooperation typically rely on one of three strategies: explicit reward engineering, in which designers manually construct reward functions that incentivize cooperative behavior; rule-based systems, where agents follow programmed ethical guidelines; or reinforcement learning with carefully designed social reward signals. While these approaches can produce functional cooperation in constrained scenarios, they suffer from fundamental limitations.

First, hand-designed reward functions often fail to generalize beyond their training conditions, resulting in agents that cooperate only in anticipated or explicitly trained situations. Second, rule-based ethical systems are inherently rigid and unable to adapt to the nuanced, context-dependent decisions required in complex social environments. Third, behaviors learned through standard reinforcement learning can be brittle, with agents discovering strategies that technically maximize rewards while violating the intended cooperative behavior. Most critically, none of these approaches produce agents with an intrinsic motivation for prosocial behavior; cooperation remains instrumental rather than genuine, existing only as long as external incentives maintain it.

These limitations highlight the need for alternative approaches that can generate intrinsic motivations for cooperative behavior rather than relying solely on external reward signals or predefined rules. Biological systems provide a compelling example of such intrinsic motivations. In many species, maternal caregiving behavior emerges from internal physiological regulation, hormonal modulation, and neural decision mechanisms shaped by evolutionary pressures. These mechanisms produce flexible and context-dependent caregiving responses without requiring explicit reward optimization.

Despite extensive research in multi-agent cooperation, relatively little work has explored biologically grounded architectures that integrate internal physiological regulation, motivational decision systems, and evolutionary dynamics within artificial agents. Most existing computational models treat agents as abstract decision-makers optimized through reward functions rather than as organisms whose behavior emerges from internal regulatory processes interacting with environmental pressures. Consequently, the minimal mechanisms required for intrinsic prosocial motivation in artificial systems remain poorly understood.

1.2.2 The Biological Solution

In contrast, biological systems have evolved robust and adaptive prosocial behaviors through millions of years of evolutionary selection. Maternal caregiving represents a particularly powerful example: mothers across many species exhibit flexible strategies that balance self-interest with offspring welfare, extend care to non-kin under certain conditions, and adapt their behavior dynamically to environmental pressures such as resource scarcity and predation risk. These behaviors emerge not from explicit rules but from the interaction of genetic predispositions, hormonal modulation, neural decision circuits, and ecological pressures acting across evolutionary timescales.

This biological perspective suggests a fundamentally different paradigm for creating prosocial artificial agents. Rather than prescribing cooperative behavior through explicit engineering, cooperation may be allowed to emerge through evolutionary mechanisms analogous to those observed in nature. If agents possess heritable variation in behavioral tendencies and experience environmental selection pressures, adaptive prosocial strategies may arise spontaneously without the need for manually designed reward functions or rigid ethical rules.

However, translating these biological principles into computational systems remains a significant challenge. Biological maternal behavior arises from complex interactions between physiological regulation, neural decision processes, and evolutionary adaptation. Designing artificial agents that capture these mechanisms in a simplified but meaningful way requires new computational frameworks capable of integrating internal state regulation, motivational decision systems, and evolutionary dynamics.

This thesis investigates such a framework through an evolutionary agent-based simulation of maternal caregiving behavior. Artificial maternal agents maintain internal physiological states and hormone-like signals that influence a motivational decision system governing caregiving, foraging, protection, and self-maintenance behaviors. Over longer evolutionary timescales, inherited motivation weights are adapted via a single-lineage search (mutation and accept-if-better selection based on mean fitness across multiple stochastic rollouts), allowing behavioral parameters to improve across generations. Through this interaction between internal regulation, plasticity, and evolutionary selection, the model explores how maternal-like caregiving strategies may emerge under varying environmental conditions.

1.2.3 Research Questions

This thesis addresses the following research questions:

Central Research Question:

How can biologically inspired internal regulatory mechanisms—such as homeostatic states, hormone-like signals, and competing motivational drives—produce adaptive maternal caregiving behavior in artificial agents without relying on explicit reward engineering?

Supporting Research Questions:

1. Biological Mechanisms:

- How do internal physiological states (such as energy and fatigue) and hormone-like signals (such as oxytocin and cortisol) influence maternal decision making?
- How do interactions between bonding, fear, and stress affect caregiving behavior?
- What minimal set of internal regulatory mechanisms is necessary to produce stable maternal caregiving strategies in artificial agents?

2. Environmental Conditions (within a fixed dyadic setting):

- How do food availability (food spawn interval) and threat difficulty (number of threats) change the balance among Forage, Care, Self, and Protect?
- Do evolved strategies generalize to unseen environment conditions (E4 transfer: world difficulty $\times$ food interval)?

3. Evolutionary and Plasticity Dynamics:

- Do caregiving-related weights and behavioral signatures reliably improve from anti-maternal and random initializations under selection?
- How do plasticity settings (E2/P1/P2) change the speed and stability of improvement?

4. Computational Framework:

- How can biological maternal care mechanisms—such as hormonal modulation, internal regulation, and evolutionary selection—be meaningfully abstracted as computational agent parameters?
- Which modeling assumptions are necessary to capture the essential dynamics of maternal caregiving behavior?
- Can insights from evolutionary maternal care models inform the design of prosocial artificial agents?

1.3 Research Objectives

1. To implement a biologically inspired maternal agent architecture with internal physiological states (energy, fatigue), hormone-like modulators (oxytocin, cortisol), and psychological states (bonding, fear, stress, closeness) driving motivational competition.

2. To study whether adaptive caregiving and protection can emerge from weighted internal regulation rather than explicit reward engineering, using the argmax selection of motivations (Forage, Care, Self, Protect).
3. To evolve inherited motivation weights through a single-lineage evolutionary search (mutation + accept-if-better selection), using fitness tied to child and mother survival outcomes with a small injury penalty.
4. To test robustness and generalization by evaluating evolved genomes across unseen environment conditions (E4: world difficulty $\times$ food interval) and comparing across plasticity regimes (E2/P1/P2).

1.4 Scope and Limitations

1.4.1 Scope (what is implemented and evaluated)

This thesis studies a minimal dyadic setting: a single mother–child pair embedded in a discrete grid-world with food resources and optional threats. Maternal behavior emerges from (i) internal state regulation (energy, fatigue; bonding, fear, stress, closeness) and two hormone-like modulators (oxytocin and cortisol), (ii) motivational competition among four drives (Forage, Care, Self, Protect) with argmax action selection, and (iii) adaptation across time scales via plasticity (within lifetime) and evolutionary selection (across generations).

1.4.2 Experimental design (what is compared)

We compare three initialization regimes (anti-maternal, random, pro-maternal) and three plasticity regimes (E2: evolution-only, P1: plastic during evolution, P2: evolved genome with runtime plasticity). Robustness is tested via E4 transfer: nine unseen conditions formed by world difficulty {easy, normal, hard} and food interval {5, 15, 25}. Each evaluation uses multiple stochastic rollouts (different random seeds) and reports median/IQR trends across evolved genomes.

1.4.3 Measurements (what we report)

The primary outcomes are normalized time-to-death (child and mother), child injury, conditional action probabilities (e.g., $P(\text{Forage} \mid \text{hungry})$, $P(\text{Care} \mid \text{cold})$, $P(\text{Protect} \mid \text{injured})$), evolved motivation weights, and child death-cause mixtures (hunger/cold/injury/alive) for interpretability.

1.4.4 Limitations (what is not modeled)

The current implementation does not include kin recognition, multiple families in the same world, explicit alloparental care, communication, or co-evolution of child signaling. Evolution is implemented as a single-lineage accept-if-better search rather than a population genetic model. These limitations are discussed in Chapter 5 as opportunities for future work.

1.5 Significance and Expected Contributions

This research addresses fundamental questions at the intersection of evolutionary biology, artificial intelligence, and multi-agent systems. By investigating how maternal caregiving behavior can emerge through biologically inspired internal regulation and evolutionary processes, this study contributes to the understanding of how intrinsic prosocial behavior may arise in artificial agents without explicit reward engineering or rigid rule-based control.

The expected contributions of this thesis include several areas:

Theoretical Contribution: This research provides a conceptual framework linking biological maternal caregiving mechanisms with computational agent architectures. By integrating internal physiological regulation, hormone-like signals, motivational decision systems, and evolutionary adaptation, the study contributes to theoretical understanding of how complex prosocial behaviors may emerge from relatively simple internal mechanisms.

Methodological Contribution: The thesis develops an evolutionary agent-based simulation model that combines internal state regulation, motivational competition, behavioral plasticity, and generational selection. This modeling framework provides a platform for investigating how caregiving behavior evolves under varying environmental conditions.

Empirical Contribution: Through systematic simulation experiments, the research identifies how food scarcity and threat difficulty shape the emergence and failure modes of caregiving strategies, and how plasticity modulates the evolutionary trajectory and generalization to unseen conditions.

Design Implications for Artificial Intelligence: The findings of this work offer biologically inspired design principles for developing artificial agents capable of adaptive prosocial behavior. Such principles may inform future research on cooperative AI systems, social robotics, and AI safety by exploring mechanisms through which intrinsic motivations for caregiving and cooperation may arise.

1.6 Thesis Organization

The remainder of this thesis is organized as follows.

Chapter 2 reviews relevant literature on maternal behavior in biology, the neurobiological mechanisms underlying caregiving, evolutionary theories of altruism, and computational models of cooperation. This chapter establishes the theoretical foundation for the proposed framework.

Chapter 3 describes the research methodology, including the maternal agent architecture, internal physiological state modeling, motivational decision mechanisms, evolutionary simulation design, and the mapping of biological principles to computational parameters.

Chapter 4 presents the experimental results and analyses, examining how different environmental pressures influence the emergence and stability of maternal caregiving strategies in artificial agents.

Chapter 5 concludes the thesis by summarizing the main findings, discussing their implications for biologically inspired artificial intelligence and AI safety, and outlining directions for future research.

Chapter 2 Literature Review

Maternal care behavior exhibits remarkable diversity across species, ranging from exclusive care for genetic offspring (kin selection) to cooperative breeding where multiple adults raise young collectively (alloparental care) to competitive territoriality where mothers defend resources aggressively. These strategies emerge from the interaction of genetic predispositions, hormonal modulation, neural decision circuits, and environmental pressures operating across evolutionary timescales. Understanding which biological mechanisms and ecological conditions favor each strategy has implications for both evolutionary biology and artificial intelligence.

Scope note (alignment with this thesis). The implementation studied in this thesis focuses on a minimal dyadic setting: a single mother–child pair in a grid-world with food resources and optional threats. The model does *not* implement kin recognition, multi-family cooperative breeding, or explicit alloparental care. We review these richer biological strategies to motivate the design space and to justify future extensions (e.g., kinship, multiple families, asynchronous worlds), while the present experiments test whether caregiving can emerge from internal regulation and evolutionary selection in the simplest setting where it is measurable.

Positioning relative to reward shaping and reinforcement learning. Many computational approaches to prosocial agents rely on externally specified objectives: reward shaping, social reward terms, or constraints that directly encode the desired cooperative outcome. In contrast, the approach in this thesis emphasizes *internal motivation* as the driver of behavior: actions are selected via competition among biologically inspired drives (Forage, Care, Self, Protect) computed from physiological and hormone-like state variables. Adaptation occurs by adjusting how strongly internal signals influence these motivations, through lifetime plasticity and evolutionary selection on inherited motivation weights. This framing aims to produce caregiving as an emergent strategy under selection pressure rather than as a behavior explicitly optimized by a hand-designed reward function.

In biological systems, maternal behavior is supported by neuro-hormonal mechanisms such as oxytocin and prolactin, and by neural circuits including the hypothalamus, amygdala, and prefrontal regions. In our computational abstraction, we model a subset of these mechanisms: oxytocin-like and cortisol-like modulators, combined with physiological and psychological state variables, drive motivational competition among Forage, Care, Self, and Protect behaviors.

2.1 Maternal Behavior Strategies in Nature

Mechanisms of Kin Recognition

Accurate kin recognition is fundamental for kin-selective maternal behaviour, as it ensures that costly care is directed toward genetically related offspring who provide inclusive-fitness returns. Across taxa, several complementary mechanisms have evolved to achieve this discrimination.

Relevance to this thesis. In the current simulation, mothers are assigned a single child and effectively have perfect identification of that child; kin recognition errors and decisions between multiple offspring are not modeled. We include kin-recognition mechanisms here as biological grounding and as a clear direction for future multi-family extensions.

Olfactory and chemical cues. In many mammals, olfactory signals are the primary substrate for kin recognition. Major histocompatibility complex (MHC)–linked odour cues allow individuals to assess relatedness through scent, as demonstrated in mice, primates, and humans. Mothers can identify their own offspring within minutes of birth through a combination of amniotic and individual odour signatures, thereby preventing mis-directed care and infanticide.

Neural and hormonal modulation. Neural circuits in the hypothalamus, amygdala, and medial preoptic area integrate sensory information with hormonal states to bias maternal responses toward familiar kin. Oxytocin and prolactin act as neuromodulators enhancing affiliative and protective behaviours once kin are recognized. In rodents, lesions in these circuits disrupt selective nursing, while oxytocin administration reinstates kin discrimination.

Social and spatial heuristics. When direct recognition is unreliable, many species rely on contextual cues such as nest location, co-rearing, or association history. Familiarity and shared developmental environment act as low-cost heuristics for kinship. In social species such as primates or elephants, social bonding and grooming networks also reflect underlying kin structure, maintaining cooperative alliances among maternal relatives.

Together, these mechanisms illustrate how kin recognition is not a single sensory process but a multi-layered integration of olfactory, neural, and social information supporting inclusive maternal strategies.

Adaptive Context

Kin-selective maternal care evolves within a broader adaptive landscape shaped by ecological constraints and genetic conflicts. While caring for kin can enhance inclusive fitness, it often entails opportunity costs—time, energy, and risk—that can reduce a mother’s residual reproductive value. Parent–offspring conflict theory predicts that offspring demand more investment than mothers are selected to give, leading to negotiation dynamics observable in feeding rates and weaning resistance [5]. Furthermore, local resource competition among relatives can offset the benefits of helping, producing evolutionary equilibria where cooperation and competition coexist.

At the genomic level, imprinting mechanisms can mediate these conflicts. Paternally expressed alleles may favor increased maternal provisioning, while maternally expressed

alleles counteract over-exploitation, producing a molecular record of evolutionary negotiation [6]. Thus, kin selection interacts with ecological and genetic pressures to produce the observed diversity of maternal strategies across species.

2.1.1 Cooperative Breeding and Alloparental Care

Defining Cooperative Breeding

Cooperative breeding refers to a social system in which individuals beyond the genetic parents assist in the care and rearing of offspring [7, 8]. These helpers—often older siblings, kin, or occasionally unrelated individuals—contribute food, protection, and social support to developing young. The system thus represents an evolutionary extension of maternal behaviour, redistributing parental effort across group members. In birds, meerkats, and some primates, cooperative breeding has been shown to enhance offspring survival under conditions of environmental uncertainty or high predation risk.

From an evolutionary standpoint, cooperative breeding can arise through multiple pathways. When helpers are related to the offspring, their participation may be explained by kin selection; when unrelated, mutualism, reciprocity, or group-level benefits may provide the fitness incentives [9]. Consequently, cooperative breeding is often viewed as a bridge between kin-biased maternal care and generalized prosocial behaviour.

Environmental and Social Determinants

Cooperative breeding tends to evolve under environmental conditions where the cost of independent breeding is high and the benefit of shared care is substantial. Key determinants include:

- **Habitat saturation:** Limited breeding sites or territories constrain dispersal and favour staying to help relatives.
- **Predation pressure:** Group vigilance and collective defense increase offspring survival in dangerous environments.
- **Resource unpredictability:** Variable food availability makes cooperative provisioning advantageous by buffering young against starvation.
- **Social stability:** Long-term bonds and dominance hierarchies permit reliable cooperation among group members.

Physiologically, hormonal modulation (e.g., prolactin elevation) has been linked to the expression of helping behaviour, providing a proximate mechanism by which environmental cues trigger cooperative responses.

2.1.2 Competitive and Territorial Maternal Strategies

Aggression, infanticide and exclusive space

While kin-selective and cooperative systems emphasise prosociality, many species exhibit maternal strategies shaped by competition rather than cooperation. Competitive maternal behaviour involves the active defense of territory, mates, or offspring and may include

aggression toward conspecific females or even infanticide of unrelated young [12, 13]. These strategies emerge when resources essential to reproduction—such as food, shelter, or male assistance—are limiting, making direct competition more profitable than sharing.

Among mammals, lactating females of species such as lions, langurs, and rodents have been observed to commit infanticide against rival offspring, thereby eliminating competitors for milk or social rank. In rodents, stress-induced suppression of lactation and nest defense behaviour are mediated by neural circuits in the amygdala and hypothalamus that are distinct from those governing affiliative care. Hormones such as progesterone, corticosterone, and testosterone interact with maternal circuits to switch behavioural priorities from caregiving to defense under conditions of threat or crowding.

Territorial exclusivity further reinforces this competitive dimension: mothers that secure private nesting or denning areas reduce the risk of alloparental interference and predation. In solitary carnivores such as leopards and bears, maternal range size and aggression levels correlate strongly with offspring survival and access to prey resources [14]. Thus, competitive maternal care can be adaptive when the ecological payoffs of exclusivity exceed the benefits of cooperation.

Resource Competition and Dominance

Dominance hierarchies modulate access to reproductive resources and shape patterns of maternal investment within social groups [15, 16, 17]. In many primates, dominant females reproduce more frequently and experience higher infant survival, while subordinates may delay breeding or engage in covert infanticide. Such rank-related asymmetries generate reproductive skew, a phenomenon explained by concession and suppression models in social evolution theory .

The neuroendocrine basis of dominance involves elevated basal glucocorticoid and androgen levels that enhance competitive motivation and vigilance [17]. From an evolutionary perspective, maternal aggression and dominance can be interpreted as conditional strategies—activated by local resource competition and demographic density rather than fixed traits. These strategies ensure that limited resources are preferentially channelled to the female’s own genetic lineage [18].

Conditions Favoring Competitive Strategies

Comparative analyses suggest that competitive or territorial maternal systems are favoured when [19, 20]:

- **Resources are clumped and defensible**, allowing individuals to exclude others profitably.
- **Predation pressure is moderate**, so group defense offers limited advantage.
- **Population density is high**, increasing direct competition among mothers for breeding sites.
- **Relatedness among local females is low**, reducing the indirect-fitness value of tolerance or cooperation.

Under these ecological and social circumstances, selection can favour aggressive maternal phenotypes, even at the cost of group cohesion. The resulting behavioural diversity—ranging from cooperative breeders to infanticidal competitors—illustrates the adaptive flexibility of maternal care strategies [13].

2.2 Biological Mechanisms of Maternal Behavior

Maternal care is underpinned by complex genetic, hormonal, and neural systems that have evolved to ensure offspring survival and optimize parental investment. While the behavioral strategies reviewed above describe how maternal care varies across ecological and social contexts, its expression ultimately depends on internal biological mechanisms that regulate perception, motivation, and decision-making. The following subsections summarize the key genetic, hormonal, and neurocircuit-level processes that mediate maternal behaviour across species.

2.2.1 Genetic and Evolutionary Basis

At the genetic level, maternal behaviour reflects both conserved and rapidly evolving components. Comparative genomics and quantitative trait analyses reveal that genes involved in social bonding, neuropeptide signalling, and stress regulation show strong signatures of selection in mammals [25, 26]. Variants in genes coding for oxytocin (OXT), vasopressin (AVP), prolactin (PRL), and their receptors (OXTR, AVPR1a, PRLR) are associated with species differences in pair bonding, aggression, and parental care.

Epigenetic mechanisms provide an additional layer of maternal adaptation. Early-life maternal care alters offspring DNA methylation patterns in genes regulating glucocorticoid receptors and stress reactivity, producing transgenerational effects on caregiving propensity [27, 28]. These findings suggest that maternal behaviour is both heritable and developmentally plastic—shaped by genetic architecture but modifiable through experience and environment.

From an evolutionary perspective, the genomic basis of care often reflects coadaptation between mothers and offspring. Genomic imprinting—where alleles are expressed in a parent-of-origin-specific manner—illustrates intragenomic conflict over maternal resource allocation [6]. Such imprinted loci (e.g., *Igf2*, *Peg3*) influence offspring growth and maternal nurturing intensity, linking molecular evolution to behavioural outcomes.

2.2.2 Hormonal Modulation

Hormonal systems coordinate the timing and intensity of maternal behaviours by coupling physiological state to environmental and social cues. The neuropeptide oxytocin plays a central role in facilitating maternal bonding, milk ejection, and affiliative behaviour across mammals [29, 30]. Its release during parturition and nursing enhances neural responsiveness to offspring cues in regions such as the amygdala and nucleus accumbens.

Prolactin, produced by the anterior pituitary, promotes maternal motivation and nest-building, while estrogen and progesterone modulate the transition from pregnancy to postpartum caregiving. Elevated estrogen increases oxytocin receptor expression, priming neural circuits for caregiving, whereas the postpartum decline in progesterone removes inhibitory control on maternal responsiveness [31].

Corticosteroids and catecholamines further shape maternal vigilance and defensive behaviour under stress. The dynamic interplay between oxytocinergic and stress-axis signalling provides a physiological substrate for flexible caregiving—allowing mothers to switch between nurturing and protective states depending on context.

2.2.3 Neural Circuits and Decision Making

Maternal behaviour arises from an integrated network of brain regions collectively termed the “maternal circuit.” Core components include the medial preoptic area (MPOA) of the hypothalamus, amygdala, nucleus accumbens, and prefrontal cortex [32, 25]. The MPOA acts as a hub integrating hormonal and sensory inputs; lesions in this region abolish maternal care, whereas stimulation reinstates it even in nulliparous females.

Sensory processing of offspring cues—olfactory, auditory, and visual—is transmitted to the amygdala, where emotional salience is assigned, and to the reward system via dopaminergic pathways. The mesolimbic dopamine system, particularly projections from the ventral tegmental area to the nucleus accumbens, mediates the motivational aspects of maternal care and reinforces caregiving behaviour through reward prediction.

Higher-order regulation occurs in the prefrontal cortex, which supports flexible decision-making and the integration of competing drives such as care, defense, and self-maintenance. Functional imaging in humans and electrophysiological studies in rodents show overlapping activation in parental and reward circuits, suggesting a shared neural architecture for caregiving and prosocial motivation.

Computationally, these neural circuits can be represented as hierarchical decision systems, where hormonal state sets gain parameters for motivational weighting, and sensory input updates action values through reinforcement-like processes. Such abstractions guide the development of artificial agents capable of adaptive caregiving decisions driven by internal states and environmental contingencies.

2.3 Environmental and Social Factors

Maternal strategies are not expressed in isolation but emerge from the interaction between intrinsic biological mechanisms and extrinsic ecological pressures. Environmental variability, population density, predation risk, and social organization all modulate the costs and benefits of caregiving. The following subsections review how resource distribution, predation, and social structure influence the evolution and expression of maternal behaviour across species.

2.3.1 Resource Availability and Competition

Resource abundance and distribution fundamentally determine the energetic feasibility and competitive dynamics of maternal care. When resources are plentiful and evenly distributed, females can invest more consistently in their offspring, leading to stable or cooperative maternal systems. Conversely, when resources are scarce or patchy, mothers must allocate care strategically, often at the expense of other females or offspring [19, 33].

In mammals and birds, food limitation has been shown to increase infanticide, abandonment, and reduced provisioning rates, particularly under high population density [13].

Resource competition also interacts with social dominance: high-ranking females in primates and carnivores monopolize feeding territories and thus produce heavier, more viable offspring [16]. In contrast, in cooperative breeders such as meerkats, shared provisioning buffers environmental variability, reducing the direct cost of scarcity per individual helper [7].

Experimental manipulations in rodents and birds demonstrate that mothers dynamically adjust litter size, nursing duration, and foraging effort in response to perceived food availability—behavioural plasticity that increases reproductive efficiency across fluctuating environments. Computationally, such adaptive allocation can be modelled as dynamic optimization under energetic constraints, where agents update maternal effort according to real-time resource feedback.

2.3.2 Predation Pressure and Group defense

Predation risk exerts a strong selective pressure on maternal behaviour and social organization. In species facing high predator density, collective vigilance and cooperative defense can significantly improve offspring survival [34, 35]. Group-living mothers often synchronize reproduction, share vigilance duties, and employ communal nesting to reduce per capita predation risk. For example, ground squirrels and meerkats emit alarm calls that alert both kin and non-kin, reflecting an adaptive trade-off between self-risk and inclusive benefits.

Predation also shapes spatial and temporal aspects of maternal care. In high-risk habitats, females exhibit more cryptic nesting behaviour, shorter foraging excursions, and enhanced aggression toward intruders. Comparative analyses in birds reveal that nest concealment and coordinated mobbing evolve most frequently in environments with intermediate predation intensity—balancing the costs of defense with reproductive payoff.

At the computational level, predation risk can be represented as an environmental hazard function influencing maternal decision policies. Agents exposed to higher simulated predation parameters may adopt risk-sensitive strategies such as shorter foraging intervals or increased clustering, reproducing the emergent trade-offs observed in nature.

2.3.3 Relatedness and social structure

Social organization defines the network through which maternal effort, protection, and learning are distributed. In species with strong kin networks, such as elephants, primates, and cetaceans, cooperative allomaternal care enhances offspring survival and accelerates social learning [36, 2]. In contrast, in low-relatedness or fluid social systems—such as solitary carnivores—maternal care remains exclusive and fiercely territorial.

The degree of relatedness among group members shapes the evolution of both cooperation and competition. High relatedness promotes tolerance, communal nesting, and shared lactation, whereas low relatedness favours individualistic or aggressive maternal tactics [20]. Long-term studies in social mammals show that stable matrilineal hierarchies enable the inheritance of social rank and knowledge, fostering intergenerational continuity of care.

At a theoretical level, social structure can be formalized as a network graph in which nodes represent individuals and edges represent relatedness or affiliative strength. Network simulations and multi-agent reinforcement learning have demonstrated that increas-

ing the weight of kinship edges enhances global stability and cooperation, echoing empirical findings in group-living species [24]. Such computational analogues provide a bridge between the statistical structure of animal societies and the emergent dynamics of artificial multi-agent systems.

2.4 Evolutionary Theories of Altruism

Altruism—defined as behaviour that benefits others at a cost to the actor—poses a fundamental challenge to classical Darwinian selection. Over the past half-century, multiple theoretical frameworks have been developed to explain how such prosocial traits can evolve and persist. These include kin selection and inclusive fitness theory, reciprocal altruism, and group or multi-level selection. Together, these models provide complementary perspectives for understanding the evolution of maternal care, cooperation, and prosociality in both biological and artificial systems.

2.4.1 Kin Selection and Inclusive Fitness

Kin selection theory explains altruism through genetic relatedness: a behaviour is favoured when the inclusive fitness benefits to relatives, weighted by relatedness, exceed the actor’s personal cost [2, 3]. This principle, formalized as Hamilton’s rule ($rb > c$), integrates direct and indirect reproductive success into a single measure of evolutionary payoff.

Empirical studies across taxa support kin selection as a central mechanism underlying maternal care and cooperative breeding. Examples include kin-biased adoption in red squirrels, sex allocation in eusocial Hymenoptera, and kin-dependent provisioning in birds and mammals [4, 7]. Inclusive fitness theory remains the dominant framework for explaining how natural selection operates on social interactions within families, linking genetic relatedness to behavioural outcomes.

Computationally, kin selection has been represented through agent-based models where relatedness functions as a probability of shared genes or mutual reward. Reinforcement-learning agents using relatedness-weighted payoff functions reproduce the emergence of helping behaviour and parental investment when the expected benefit-to-cost ratio satisfies Hamilton’s inequality. These models provide formal bridges between evolutionary biology and multi-agent AI [24].

Statement of Hamilton’s Rule

Hamilton’s rule gives the condition under which a social behaviour that is costly to an actor but beneficial to a recipient will be favoured by natural selection via inclusive fitness:

$$rb - c > 0, \quad (2.1)$$

where $c > 0$ is the direct fitness cost to the actor, $b > 0$ is the direct fitness benefit to the recipient, and r is the genetic relatedness between actor and recipient[2, 3].

2.4.2 Reciprocal Altruism

Reciprocal altruism expands the scope of cooperation beyond kin by allowing repeated interactions and conditional strategies [37]. In this framework, individuals help others

with the expectation of future return, creating a feedback loop that stabilizes cooperation over time. Iterated game-theoretic models such as the “tit-for-tat” strategy demonstrate that reciprocity can evolve when individuals repeatedly interact and can remember partners’ previous behaviours [38].

Empirical support for reciprocity is found in vampire bats that share blood meals with previous donors, cleaner fish that maintain mutualistic relationships with clients, and primates that exchange grooming for coalitionary support [39, 40]. In maternal contexts, reciprocal care is observed in communal nesting rodents and cooperative primates, where mothers alternately nurse or guard each other’s offspring, generating long-term fitness gains for all participants.

In computational modelling, reciprocal altruism is captured through reinforcement learning with memory or state-dependent cooperation. Agents adopt conditional helping policies that reinforce cooperation when interactions are repeated and partner identity is retained. These simulations illustrate how reciprocity and reputation tracking can sustain altruistic dynamics even in the absence of genetic relatedness.

2.4.3 Group Selection and Multi-level Selection

Group selection, and its modern extension—multi-level selection theory—proposes that natural selection acts not only at the level of individuals or genes but also among groups. Cooperative traits that are costly within groups can nevertheless spread if they enhance the survival or reproduction of the group as a whole [41, 42]. In this view, individual fitness is nested within group fitness, and evolutionary stability depends on the relative strength of selection at each level.

Empirical examples include eusocial insects, cooperative birds, and human societies, where groups with higher internal cooperation outcompete less cohesive ones. Mathematical formulations of multi-level selection are formally equivalent to inclusive fitness models under many conditions, though they emphasize different causal partitions of selection [43].

In computational and robotics contexts, multi-level selection is implemented through population-level optimization or hierarchical reinforcement learning. Agents organized into collectives are rewarded for both individual and group performance, leading to emergent division of labour and cooperative specialization. These models parallel the evolution of task differentiation and social cohesion observed in biological systems.

Collectively, these theories—kin selection, reciprocity, and multi-level selection—illustrate how altruism can evolve through different but overlapping pathways. For maternal care, they provide an integrated explanatory framework: kin selection accounts for family-directed investment, reciprocity captures cooperative exchanges among peers, and multi-level selection explains the emergence of group-level coordination. Together, they inform both the biological understanding of caregiving and the design of prosocial, bio-inspired artificial agents.

2.5 Computational Models and Bio-inspired Robotics

Computational modelling provides a quantitative and mechanistic bridge between evolutionary theory and the design of intelligent agents capable of social and caregiving be-

haviour. By abstracting the principles of kin selection, reciprocity, and cooperation into formal decision rules, computational models allow systematic exploration of how altruistic and maternal-like behaviours can emerge in artificial systems. This section reviews three major modelling paradigms—game theory and agent-based simulations, reinforcement learning and emergent cooperation, and bio-inspired applications to caregiving and prosocial AI.

2.5.1 Game theory and agent-based models

Evolutionary game theory provides a foundational framework for analysing the conditions under which cooperation and altruism evolve. Early models such as the Prisoner’s Dilemma and Hawk–Dove games demonstrated that simple payoff structures can capture complex trade-offs between competition and cooperation [21, 22]. Within these frameworks, strategies evolve according to their relative fitness payoffs, revealing equilibrium points such as Evolutionarily Stable Strategies (ESS).

Agent-based models (ABMs) extend game-theoretic reasoning by simulating populations of autonomous individuals who interact locally under ecological constraints. These models can incorporate relatedness, spatial proximity, resource availability, and recognition error, allowing direct testing of Hamiltonian and group-selection predictions in dynamic environments [44, 45]. For example, agents governed by the rule $rb > c$ spontaneously form kin clusters and cooperative alliances when migration rates and recognition accuracy are tuned appropriately.

ABMs have also been used to study the evolution of parental and alloparental care, demonstrating how cooperation can emerge from repeated local interactions and feedback between social structure and resource ecology. Such simulations provide a conceptual bridge between animal behaviour and synthetic populations of robotic or digital agents, where cooperation arises from decentralized decision rules rather than centralized control.

2.5.2 Reinforcement learning and emergent cooperation

Reinforcement learning (RL) extends evolutionary principles to adaptive agents that learn through experience rather than genetic inheritance. In multi-agent RL (MARL), cooperation and altruism emerge as agents discover policies that maximize cumulative reward within shared environments [23, 24]. When reward functions include components for collective or partner success, agents learn prosocial strategies analogous to biological cooperation and parental care.

Recent work demonstrates that RL agents equipped with memory, empathy-like state estimation, or communication channels can evolve reciprocity, fairness, and resource sharing—behaviours long studied in evolutionary biology. For instance, agents trained in sequential social dilemmas display adaptive alternation between competitive and cooperative phases, reminiscent of conditional altruism observed in primates and cooperative breeders [46, 47].

From a neural perspective, RL architectures parallel the biological reinforcement systems that guide caregiving behaviour. Dopaminergic-like reward prediction mechanisms, state-value representations, and hormonal modulation analogues can be implemented to reproduce the dynamic balance between nurturing, defense, and self-maintenance. This

convergence highlights the shared computational logic between biological and artificial systems of social learning.

2.5.3 Applications to caregiving and prosocial AI

Insights from biological maternal behaviour and cooperation have inspired the emerging field of bio-inspired robotics and prosocial AI. In robotics, caregiving algorithms have been implemented to enable companion or assistive robots to prioritize vulnerable agents, maintain proximity, and provide adaptive aid [48, 49]. These systems often employ emotional appraisal modules or hormonal analogues (e.g., “oxytocin-like” control parameters) to regulate social engagement intensity.

At the population level, swarm and collective robotics use multi-agent cooperation to achieve group-level resilience and resource allocation reminiscent of cooperative breeding systems [50, 51]. Here, altruistic or defensive behaviours can be modelled through reward shaping and hierarchical coordination, producing emergent task specialization analogous to maternal defense or communal care.

In prosocial AI, evolutionary and reinforcement-based models are now being combined with neural architectures that incorporate empathy, reciprocity, and fairness constraints. These approaches aim to build artificial agents capable of context-sensitive altruism—agents that balance self-interest with concern for others’ welfare [24, 1]. Integrating these biologically grounded principles into AI design not only advances robotics but also offers a framework for embedding ethical and adaptive caregiving behaviours into future intelligent systems.

2.5.4 Neuroplasticity, Hebbian learning, and homeostasis (conceptual links)

Biological caregiving is shaped not only by evolutionary selection but also by within-lifetime adaptation in neural and physiological systems. **Neuroplasticity** refers broadly to the capacity of the nervous system to change its functional organization in response to experience. A classical family of mechanisms is **Hebbian learning**—often summarized as “cells that fire together wire together”—which formalizes how correlated activity can strengthen synaptic connections and thereby change behavioral tendencies over time [52].

A complementary principle is **homeostasis**: organisms regulate internal variables (energy balance, temperature, stress load) around viable ranges despite fluctuating environments. In computational terms, homeostatic regulation can be viewed as maintaining internal state variables within bounds by selecting actions that counteract deficits. This framing motivates the thesis design choice to drive behavior via internal state deficits and hormone-like modulators rather than via purely external reward shaping [53].

Relevance to this thesis. The plasticity mechanisms in this thesis are not neuron-level Hebbian synaptic learning; instead, they adjust *motivation weights* that map internal deficits and modulators to action selection. Nevertheless, the biological literature motivates the idea that *within-lifetime parameter change* can improve behavior (plasticity), while longer-term selection can stabilize beneficial settings (evolution), consistent with a Baldwin-style interaction between learning and evolution.

2.5.5 Genetic algorithms vs. evolutionary strategies (what we use)

Many artificial life and optimization studies use **genetic algorithms (GAs)**—population-based evolutionary search with selection, recombination, and mutation—as a general-purpose mechanism for discovering controllers [54, 55]. In contrast, the optimization procedure used in this thesis is a **single-lineage** (1+1) evolutionary strategy: each generation proposes one Gaussian mutation of the current best genome and accepts it only if fitness improves [56, 57]. This choice emphasizes mechanistic interpretability of a single evolving maternal lineage and reduces confounds from population dynamics, while still implementing the core evolutionary ingredients of variation and selection.

A particularly relevant bioinspired robotics study is that of Maroto-Gómez et al. (2024) [58], who proposed a neuroendocrine model for adaptive social behavior in a social robot using oxytocin, arginine vasopressin, and dopamine as internal modulators of bonding and action selection. In their architecture, perceived social stimuli influence artificial neuroendocrine variables, which then regulate physiological and psychological processes, motivations, and ultimately behavior. A central component of the model is a dynamic pair-bonding variable, which increases under positive interaction and decreases under aversive interaction, thereby shaping future social responses. Their results showed that this bioinspired pair-bonding mechanism improved perceived attachment and social presence compared with a random behavior baseline. This work is especially relevant to the present thesis because it demonstrates that biologically inspired internal regulation can generate adaptive social behavior in artificial agents. However, its focus is human–robot interaction within a single robot-user relationship, whereas the present thesis extends the problem toward maternal caregiving, environmental pressures, and evolutionary emergence across generations.

2.6 Research Gap and Contribution

Despite extensive research across biology, neuroscience, and artificial intelligence, a critical interdisciplinary gap remains in understanding how maternal-like caregiving behavior can emerge in artificial systems. Existing work has examined the biological mechanisms of maternal care, theoretical models of altruism, and computational simulations of cooperation, yet these domains remain only loosely connected. Bridging them requires a unified framework that translates the adaptive logic of maternal behavior into computational form while remaining sufficiently simple for systematic simulation and analysis.

Recent work in bio-inspired robotics has begun to explore this direction. For example, Maroto-Gómez et al. (2024) [58] proposed a neuroendocrine-inspired architecture for social robots in which hormone-like variables such as oxytocin, arginine vasopressin, and dopamine regulate pair-bonding and behavioral responses during human–robot interaction. Their model demonstrates that internal physiological-style variables can influence social motivation and adaptive behavior in artificial agents.

However, this framework focuses primarily on pair-bonding dynamics within short-term human–robot interaction sessions. It does not address how caregiving strategies might emerge or evolve within populations of artificial agents interacting across generations under ecological constraints. As a result, the evolutionary mechanisms through which maternal-like caregiving behaviors could arise in artificial systems remain largely unexplored.

Another limitation of existing biologically inspired architectures is that many models rely on manually specified parameter values. In the work of Maroto-Gómez et al. (2024) [58], the influence of internal variables on motivational processes is determined by pre-defined weights that are selected heuristically during system design. While this approach demonstrates that neuroendocrine-inspired internal states can influence social behavior, it leaves open the question of how such parameters should arise autonomously in artificial systems.

In biological organisms, behavioral regulation parameters are not manually specified but instead emerge through a combination of evolutionary selection and adaptive plasticity during an individual’s lifetime. Inspired by this principle, the present thesis investigates whether the parameters governing maternal decision-making can emerge through evolutionary processes and plastic adaptation rather than being predetermined by the system designer.

Specifically, this research introduces two complementary mechanisms. First, plasticity mechanisms allow motivational weights to adjust during an agent’s lifetime based on behavioral outcomes. Second, evolutionary selection operates across generations, allowing inherited behavioral parameters to evolve according to offspring survival. Together, these mechanisms enable the discovery of behavioral regulation parameters through adaptive processes rather than manual specification.

This thesis therefore develops an evolutionary agent-based framework that integrates biologically inspired internal regulation, motivational decision processes, and generational selection. Through this framework, the study investigates how maternal caregiving strategies may emerge and stabilize in artificial populations under varying environmental pressures.

2.6.1 Identified Research Gap

The preceding literature reveals three important but only partially integrated research streams:

- **Biological studies of maternal care** identify genetic, hormonal, and neural mechanisms underlying caregiving behavior. However, these findings are primarily descriptive and are rarely translated into computational architectures for artificial agents.
- **Evolutionary theories of altruism**—including kin selection, reciprocity, and multi-level selection—explain why prosocial behaviors may evolve. However, they do not directly specify how internal regulatory mechanisms produce context-dependent caregiving decisions within individual agents.
- **Computational models of cooperation** in artificial intelligence and robotics demonstrate emergent coordination and adaptation. However, many such models rely on simplified agents and often do not incorporate biologically inspired internal states, motivational competition, or hormone-like modulation mechanisms.

Research gap: There is currently no widely established computational framework that simultaneously integrates (i) biologically inspired internal regulation, (ii) motivational decision mechanisms, and (iii) evolutionary adaptation across generations in order to inves-

tigate the emergence of maternal-like caregiving behavior in artificial populations. Consequently, the minimal mechanisms required for context-sensitive and intrinsically motivated caregiving behavior in artificial agents remain insufficiently understood.

2.6.2 Research Contribution

This thesis contributes to addressing this gap through the following elements:

1. **An evolutionary agent-based model of maternal caregiving behavior.** This research develops a multi-agent simulation in which maternal agents interact with offspring, environmental resources, and potential threats across multiple generations. Behavioral tendencies are represented through inheritable computational parameters that influence caregiving, protection, self-maintenance, and foraging behavior.
2. **A computational abstraction of biological maternal mechanisms.** The proposed model represents key biological concepts—including internal physiological regulation, hormone-like modulation, and motivational decision-making—as computational variables and weighted interactions that govern maternal behavior.
3. **Integration of lifetime plasticity and evolutionary adaptation.** The framework integrates behavioral adaptation within an agent’s lifetime through plasticity mechanisms together with evolutionary selection operating across generations. This dual-timescale approach allows investigation of how caregiving strategies adapt both during an agent’s lifetime and through long-term evolutionary processes.
4. **Systematic investigation of environmental determinants.** Controlled computational experiments are used to examine how environmental factors such as resource availability, environmental threats, and population structure influence the emergence and stability of maternal behavioral strategies.
5. **Design implications for prosocial artificial agents.** The resulting framework provides biologically inspired principles for constructing artificial agents capable of adaptive and context-sensitive caregiving behavior without relying exclusively on explicit reward engineering or rigid rule-based control.

2.6.3 Expected Outcomes

The research is expected to generate both scientific and computational contributions:

- **Scientific insight:** Improved understanding of how simplified biological mechanisms (internal regulation, hormone-like modulation, behavioral plasticity, and evolutionary inheritance) can give rise to adaptive caregiving behavior.
- **Computational contribution:** A simulation framework for studying the emergence of maternal-like behavior in artificial agent populations under varying environmental conditions.
- **AI design implications:** Conceptual and practical guidance for designing prosocial artificial agents whose behavior is shaped by intrinsic regulatory mechanisms and adaptive evolutionary pressures rather than solely by externally specified reward functions.

2.6.4 Summary

In summary, this thesis integrates biological insights on maternal behavior, evolutionary theory, and computational modeling within a unified framework for studying the emergence of caregiving strategies in artificial agents. By treating maternal behavior as an adaptive and evolving property of biologically inspired systems, this research contributes both to the scientific understanding of prosocial behavior and to the design of more adaptive and socially aligned artificial intelligence systems.

Chapter 3 Methodology

3.1 Overview and Research Design

This research employs a multi-method computational approach that integrates neuroendocrine-inspired behavioral mechanisms with evolutionary agent-based modeling. The methodology combines three complementary components: (i) a simplified neuroendocrine-inspired internal regulation model using oxytocin- and cortisol-like modulators, (ii) a single-lineage evolutionary search procedure for adapting inherited motivation weights across generations, and (iii) a spatial grid-world simulation for environmental interactions. The design follows a bottom-up approach where individual-level internal regulation produces measurable behavioral and survival outcomes, and selection pressure shapes inherited parameters over generations.

3.2 Conceptual Framework

3.2.1 Neuroendocrine-Behavioral Pipeline

The behavioral architecture implemented in this thesis follows a biologically inspired neuroendocrine decision pipeline. Agent behavior emerges from the interaction between environmental perception, internal physiological states, hormone-like modulators, and motivational competition.

Environment → Perception → Physiological States → Neuroendocrine Modulation → Motivational Computation → Action Selection

In addition to this behavioral loop operating at each simulation step, the model incorporates two adaptive processes that operate at different time scales:

Plasticity (within lifetime) Evolution (across generations)

Plasticity allows behavioral parameters to adjust based on experience during an agent's lifetime, while evolutionary selection propagates successful behavioral strategies across generations according to an explicit fitness function computed from stochastic rollouts (survival time and injury signals).

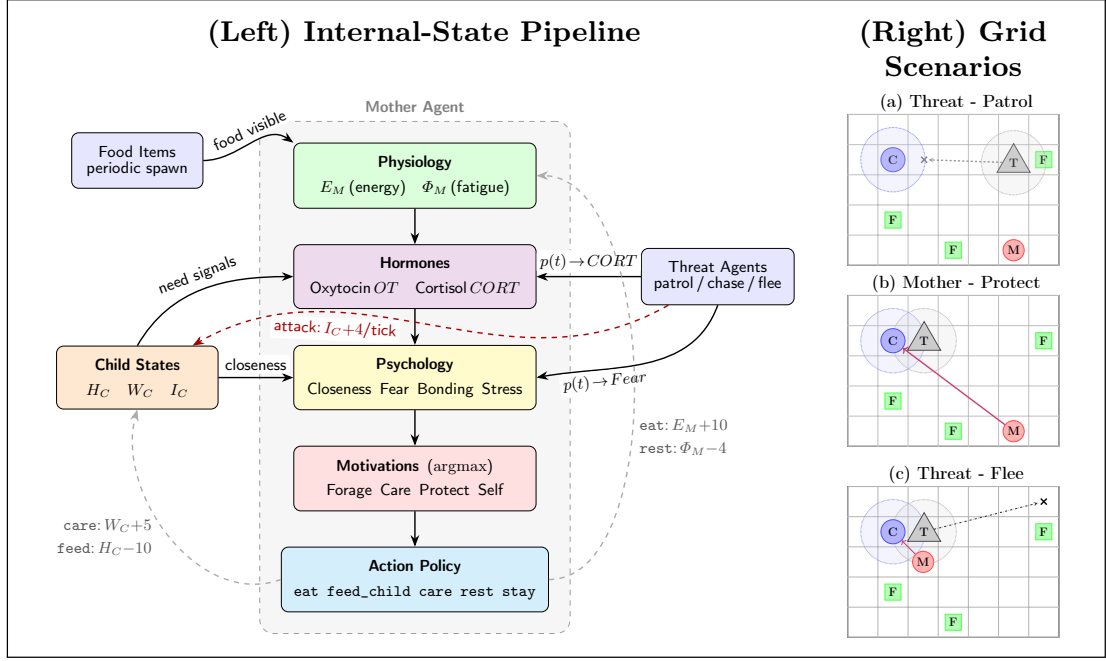


Figure 3.1: **Left:** Internal-state pipeline of the mother agent. Solid arrows indicate state/perceptual signals, dashed gray arrows indicate action feedback, and dashed red arrows indicate threat-induced child injury. **Right:** Illustrative grid-world scenarios: (a) threat patrol, (b) mother protective approach, (c) threat flee. In the grid panels, arrows indicate movement direction rather than internal signals. **C** = child, **M** = mother, **T** = threat, **F** = food.

Figure 3.1 illustrates the interaction between perception, physiological and psychological regulation, neuroendocrine modulation, and motivational decision processes in the proposed maternal agent architecture.

Genetic Traits. Genetic traits represent inherited behavioral parameters that influence how internal signals affect motivational decisions. In the implemented codebase, the evolved genome corresponds to fixed motivation weights that map internal signals (child need, energy deficit, fear/stress, bonding/closeness) into competing motivation activations. Across generations, these fixed weights are mutated and selected via an accept-if-better rule based on mean fitness over multiple rollouts.

Perception. Perception represents the mechanism through which agents detect environmental cues and transform them into internal stimuli. In the implemented model, maternal perception includes nearby food resources, threat proximity, and spatial proximity to the child. Child need is represented by child state variables (hunger, warmth, injury), which the maternal internal regulation uses as inputs to motivation computation.

Physiological States. Physiological states represent homeostatic variables describing the agent's internal condition. In the maternal agent these include energy level, fatigue, and other survival-related variables. In child agents, internal states include hunger, warmth

deviation, and injury level. These variables evolve dynamically over time and generate deficits that activate motivational processes.

Neuroendocrine States. Neuroendocrine states represent hormone-like modulatory variables that regulate social and stress-related behavior. In the maternal agent, oxytocin promotes bonding and caregiving responses, while cortisol represents stress induced by environmental threats or internal deficits. These hormone-like variables integrate information from physiological states and environmental cues, influencing motivational priorities.

Psychological (Motivational) States. Psychological states represent internal motivational drives that determine behavioral priorities. These drives emerge from the interaction between physiological deficits and neuroendocrine modulation. In the maternal agent, competing motivations include caregiving, foraging, self-maintenance, and protection. The strength of each motivation is computed from weighted combinations of internal signals, and the dominant motivation determines the selected behavior.

Motivational States. Motivational states represent competing behavioral drives generated from internal physiological and neuroendocrine signals. Each motivation is computed as a weighted integration of relevant internal signals, producing activation values corresponding to behavioral goals such as foraging, caregiving, self-maintenance, or protection. The motivation with the highest activation determines the selected behavioral tendency.

Action Output. The action output represents the behavioral decision selected by the agent at each simulation step. The agent executes the action associated with the highest motivational activation. Maternal behavior is organized by four motivations (Forage, Care, Self, Protect) and implemented through primitive interactions such as movement, eating/-collecting food, feeding the child, and staying close to provide warmth and protection. The child does not execute actions in the evaluated setting; instead, the mother observes the child's internal state variables (hunger, warmth, injury) and reacts through her motivation system. These actions modify the environment and close the perception–action feedback loop.

Adaptive Mechanisms. Two adaptive mechanisms operate in the proposed framework. First, plasticity mechanisms allow motivation-related weights to adjust during an agent's lifetime based on behavioral outcomes and internal state regulation. Second, evolutionary selection operates across generations via a single-lineage search: at each generation a candidate genome is produced by mutation, evaluated over multiple stochastic rollouts, and accepted only if mean fitness improves. Together, these mechanisms enable the discovery of behavioral regulation parameters through adaptive processes rather than manual specification.

3.2.2 Grid World Environment

The simulation environment consists of a 2D discrete grid where spatial relationships govern agent interactions. Each cell can contain agents, resources, threats, or remain empty.

The environment implements:

- **Spatial dimensions:** fixed $N \times N$ grid (default 15×15 in experiments)
- **Resources:** food items spawn periodically at random empty cells (controlled by a spawn interval)
- **Threats:** mobile threat agents that patrol/chase and can cause child injury upon close contact

World rules and key numerical constants

Table 3.1 summarizes the primary constants used in the current code implementation (state bounds, per-tick dynamics, and basic world timing). These are the values that set the time scales in the experiments and in the final presentation.

Table 3.1: Key numerical constants and world rules (implementation)

Symbol / name	Value	Meaning in code
$X_{\max}$	100	Clamp range for mother/child internal states
Day length	100 ticks	One day is defined as 100 ticks (time indexing)
Child hunger increment	+0.5/tick	Hunger increases linearly over time
Child warmth decrement	−0.5/tick	Warmth drops linearly over time (cold world)
Child injury recovery	0.5/tick	Injury recovers only when threats are far
Threat recovery radius	$0.2 \cdot \max(W, H)$ (min 1)	“Threats are far” condition for injury recovery
Basal metabolism	−0.2 energy/tick	Mother energy decreases every tick even if idle
Move cost	−0.3 energy, +0.5 fatigue	Cost applied when mother moves
Action cost	−0.1 energy, +0.1 fatigue	Cost applied when mother acts (non-move)
Rest recovery	−4.0 fatigue/tick	Fast fatigue recovery when resting
Idle recovery	−1.0 fatigue/tick	Moderate fatigue recovery when idle
Threat perception range	2 cells	Threat switches mode based on local perception

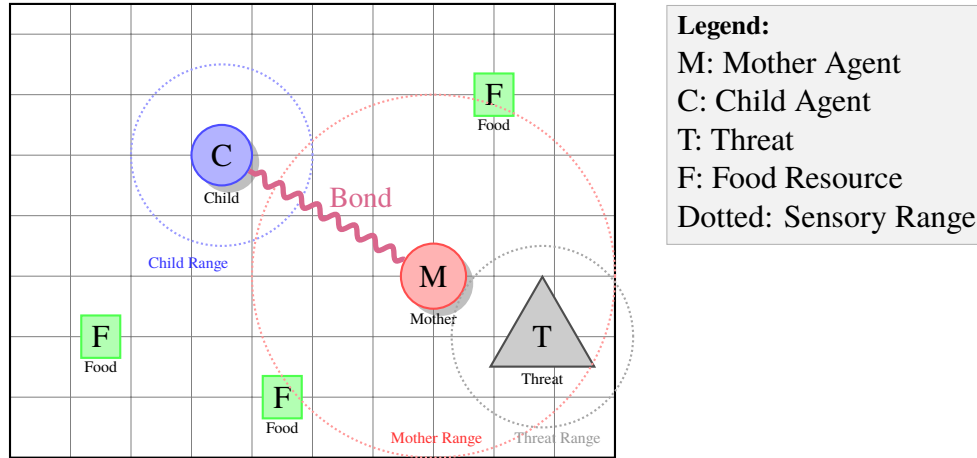


Figure 3.2: A 2D grid world representation of the **Mother-Child Interaction Model**. The mother (M) searches for food (F) and protects the child (C) within the environment while responding to nearby threats (T). Dotted circles indicate sensory perception ranges, and the wavy purple line represents the maternal bond.

The simulation environment contains three agent types and one non-agent entity, each contributing distinct functional roles within the grid-world ecosystem.

Mother Agent: The mother agent implements the full neuroendocrine decision architecture described before. It possesses complete perception of relevant environmental cues within a specified perception radius, including nearby food resources, threats, and child need variables. Based on perceived stimuli and internal physiological and neuroendocrine states, the mother evaluates competing motivations such as caregiving, protection, and self-maintenance. The selected action may include approaching the child, feeding, warming, protecting, avoiding threats, or foraging for resources.

Child Agent: The child agent represents a simplified infant model with limited perception and no independent mobility. Instead of performing complex actions, the child maintains internal physiological states including hunger, warmth deviation, and injury level. When these internal deficits increase, they influence the mother's internal regulation and motivational computation. The child therefore functions primarily as a passive vulnerable entity whose survival depends on maternal behavior and environmental conditions.

Threat Agent: Threat agents represent mobile environmental hazards. Each threat follows a random patrol movement pattern within the grid-world. Threat agents possess a limited perception radius and modify their behavior based on perceived agents. When a threat detects a child agent within its perception range, it switches from patrol to pursuit and attempts to move toward the child. Conversely, if the threat detects the mother agent, it switches to avoidance behavior and attempts to move away from the mother. This asymmetric interaction creates dynamic environmental pressure that requires protective maternal responses.

Food Entity: Food entities are passive environmental resources placed within the environment. When collected by the mother agent, food replenishes maternal energy reserves and helps maintain physiological homeostasis. Food entities do not exhibit autonomous behavior and serve only as consumable resources.

3.3 Multi-Agent System (MAS) Formulation

The simulation environment is modeled as a multi-agent system consisting of a set of interacting agents embedded in a shared spatial environment.

$$\mathcal{N} = \{a_1, a_2, \dots, a_K\}.$$

Each agent a_i is defined as:

$$a_i = \langle G_i, S_i, P_i, H_i, M_i, \pi_i, A_i \rangle$$

where

- G_i : genetic trait vector inherited across generations
- S_i : internal physiological state space
- P_i : perception function mapping environment to observations
- H_i : neuroendocrine update mechanism
- M_i : motivational computation
- π_i : behavioral policy selecting an action
- A_i : action space

The environment is represented by

$$E(t) = \langle \text{agents, resources, threats, spatial grid} \rangle.$$

At each discrete time step t , agent behavior follows a perception– regulation–action cycle.

$$\begin{aligned} o_i(t) &= P_i(E(t)) \\ s_i(t) &= S_i(t) \\ h_i(t) &= H_i(s_i(t), o_i(t), G_i) \\ m_i(t) &= M_i(s_i(t), h_i(t), o_i(t), G_i) \\ a_i(t) &= \pi_i(m_i(t), s_i(t), o_i(t)) \\ s_i(t+1) &= f_i(s_i(t), h_i(t), a_i(t)) \\ E(t+1) &= g(E(t), a_1(t), \dots, a_K(t)) \end{aligned}$$

The perception function P_i extracts environmental cues such as nearby child-need cues, food resources, and threats. These observations influence the agent’s physiological and neuroendocrine states.

The neuroendocrine update mechanism H_i models simplified hormone-like dynamics (oxytocin and cortisol) which regulate bonding, stress response, and motivational priorities.

Motivational values $m_i(t)$ are computed through the integration of physiological deficits, hormonal signals, and environmental observations. The policy function π_i selects the behavior with the highest motivational activation.

Finally, the environment transition function g updates spatial conditions and resource distributions according to all executed actions.

This perception–internal regulation–action loop forms the core behavioral mechanism of the agents.

3.4 Internal State Spaces of Agents

3.4.1 Mother Agent

The mother agent implements the full neuroendocrine architecture:

$$a_m = \langle G_m, S_m^{phys}, S_m^{psych} P_m, H_m, M_m, \pi_m, A_m \rangle$$

where

$$S_m^{phys} = \{\text{energy, fatigue}\}$$

$$S_m^{psych} = \{\text{closeness-child, bonding, fear-threat, stress}\}$$

$$H_m = \{\text{oxytocin, cortisol}\}$$

$$M_m = \{M_{\text{care}}, M_{\text{protect}}, M_{\text{forage}}, M_{\text{self}}\}$$

$$A_m = \{\text{Move/Stay, Eat, Feed child, Care (proximity-based), Protect (proximity-based)}\}$$

The mother agent evaluates environmental cues, hormonal states, and physiological deficits in order to regulate caregiving, protection, and self-maintenance behaviors.

World time and update loop (implementation)

The simulation advances in discrete ticks. One tick corresponds to one full update loop (perception $\rightarrow$ internal updates $\rightarrow$ action selection $\rightarrow$ world transition). In the default configuration, one day is defined as 100 ticks. Food spawning uses a fixed interval (explicitly configured in experiments), and threats are represented as explicit threat agents.

Physiological state dynamics (mother)

Maternal energy decreases each tick due to basal metabolism and may decrease further due to movement and actions; energy increases when the mother eats. Fatigue accumulates with movement/actions and recovers when resting or idling. These quantities are clamped to $[0, 100]$.

Hormonal and psychological meaning (mother)

Oxytocin is a proximity-driven bonding modulator (higher when the mother stays close to the child). Cortisol is a stress modulator driven by threat proximity, child need, and maternal energy deficit. Psychological states include closeness to child (inverse distance), bonding strength, fear in response to threats, and overall stress load.

3.4.2 Child Agent

Child agents implement a simplified regulatory system.

$$a_c = \langle G_c, S_c, P_c, \rangle$$

Child internal states consist primarily of need variables:

$$S_c = \{\text{hunger, warmth, injury}\}$$

The child agent has no independent mobility in the evaluated setting. Child needs are represented by state variables (hunger, warmth, injury) that the mother uses as inputs when computing motivation scores.

Baseline child dynamics without maternal interaction

All child state variables are clamped to $[0, 100]$. Hunger increases by $+0.5$ per tick from an initial value of 0. Warmth decreases by -0.5 per tick from an initial value of 50 (comfortable). Injury increases only via threat contact events and recovers when threats are far. Without any maternal interaction, the child reaches lethal cold in approximately 100 ticks (warmth $50 \rightarrow 0$), which is faster than starvation (hunger $0 \rightarrow 100$ in 200 ticks).

3.4.3 Action Spaces

Mother Agent.

$$A_m = \{\text{move, stay, care, pick food, eat, give to child}\}.$$

Child Agent. No action just signals.

Threat Agent.

$$A_{\text{threat}} = \{\text{Move, emit danger to child}\}.$$

3.5 Neuroendocrine Regulation

The internal neuroendocrine states of the mother agent evolve over discrete time steps according to stimulus-driven regulatory dynamics. Each hormone represents a simplified computational analogue of biological signaling systems involved in maternal behavior.

Let $H^{(k)}(t)$ denote the level of hormone k at time step t , where

$$k \in \{\text{OT, C}\}.$$

Hormonal dynamics follow a discrete update rule:

$$H^{(k)}(t+1) = H^{(k)}(t) + \alpha_k S^{(k)}(t) - \lambda_k H^{(k)}(t)$$

where

- $S^{(k)}(t)$ is the stimulus input derived from perception
- α_k is a stimulus sensitivity parameter
- λ_k is a decay constant

Stimulus inputs depend on environmental cues and internal physiological states perceived by the mother agent. For example:

$$S^{(OT)}(t) = f(d_c(t))$$

where d_c is the distance between the mother and the child.

Similarly, cortisol dynamics depend on threat perception and physiological stress:

$$S^{(C)}(t) = w_1 S_{threat}(t) + w_2 D_{child}(t) + w_3 D_{energy}(t)$$

where

- $S_{threat}(t)$ is perceived threat intensity
- $D_{child}(t)$ represents *child-need signals* derived from the child state variables (hunger, warmth, injury) that are directly observed by the mother
- $D_{energy}(t)$ represents maternal energy deficit

Hormone levels are constrained within the normalized range

$$H^{(k)}(t) \in [0, 1].$$

Maternal bonding strength is influenced by oxytocin and cortisol levels according to

$$MB(t) = MB(t-1) + \beta_1 OT(t) - \beta_2 CORT(t) - \delta_r \quad (3.1)$$

where MB denotes maternal bonding strength and δ_r is a natural decay rate representing weakening attachment in the absence of reinforcing interaction.

Hormonal states influence motivational priorities by modulating the activation level of competing behavioral drives.

$$M_j(t) = \sum_i u_{ji} x_i(t)$$

where

- $M_j(t)$ is the activation of motivation j
- $x_i(t)$ are physiological or hormonal signals

- u_{ji} are motivational weights

The selected behavior corresponds to the maximum activation:

$$a(t) = \arg \max_j M_j(t)$$

3.6 Evolutionary Mechanisms

Single-lineage (1+1) Evolution

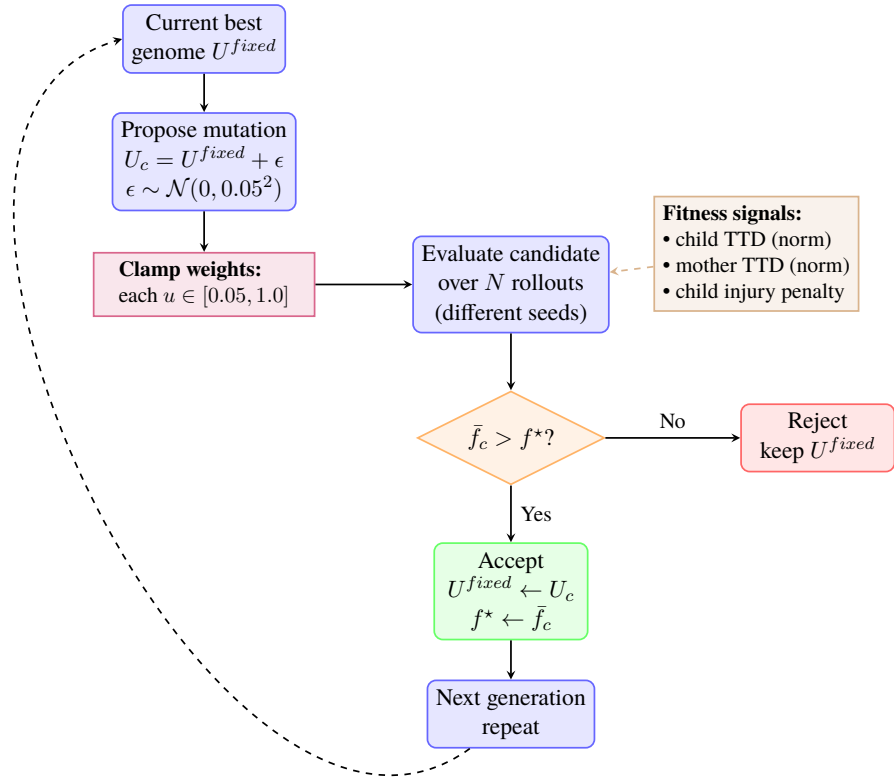


Figure 3.3: Single-lineage (1+1) evolutionary search used in this thesis. Each generation proposes one Gaussian mutation of the current best motivation-weight genome, evaluates it over N stochastic rollouts, and accepts it only if mean fitness improves.

Algorithm 1 Single-lineage (1+1) evolutionary search (implementation)

```
1: Parameters:  $G$  generations,  $N$  episodes per evaluation, mutation  $\sigma = 0.05$ 
2: Fitness:  $F = \alpha \text{TTD}_{child} + (1 - \alpha) \text{TTD}_{mother} - \lambda \text{Injury}_{child}$ 
3: Initialize best genome  $U^{\text{fixed}} \leftarrow$  baseline motivation weights (anti / random / pro)
4: best fitness  $f^* \leftarrow -\infty$ 
5: for  $g = 1$  to  $G$  do
6:   Propose candidate genome  $U_c \leftarrow U^{\text{fixed}} + \epsilon$ , where  $\epsilon \sim \mathcal{N}(0, \sigma^2)$  elementwise
7:   Clamp each weight in  $U_c$  to  $[0.05, 1.0]$ 
8:   Evaluate  $U_c$  over  $N$  stochastic rollouts (different RNG seeds) and compute mean
       fitness  $\bar{f}_c$ 
9:   if  $\bar{f}_c > f^*$  then
10:     Accept:  $U^{\text{fixed}} \leftarrow U_c$ 
11:      $f^* \leftarrow \bar{f}_c$ 
12:   else
13:     Reject: keep  $U^{\text{fixed}}$  unchanged
14:   end if
15: end for
16: return best genome  $U^{\text{fixed}}$  and trace statistics
```

3.6.1 Darwinian vs. Baldwinian in this thesis

We use the terms **Darwinian** and **Baldwinian** to distinguish two time scales of adaptation. Darwinian evolution refers to inherited fixed weights mutating across generations and being selected when they improve fitness. Baldwinian adaptation refers to within-lifetime plasticity that can improve behavior during an episode, even though plastic weight changes are not directly inherited. In our implementation, inheritance applies to fixed weights, while plastic weights reset each episode and only influence the effective policy within that lifetime.

3.6.2 Genetic Encoding

In the proposed model, each mother agent inherits a set of behavioral parameters that regulate both neuroendocrine dynamics and motivational decision processes. These parameters correspond to the fixed weights used in the agent’s internal regulatory system.

The genome of an agent is therefore represented as

$$G = \{U^{\text{fixed}}\}$$

where

- U^{fixed} represents inherited motivation weights that map internal signals (child need, energy deficit, fear/stress, bonding/closeness) into four competing motivations (Forage, Care, Self, Protect).

The motivational weight matrix

$$U^{\text{fixed}}$$

determines how physiological deficits and neuroendocrine signals contribute to the activation of competing behavioral motivations.

During the lifetime of an agent, plastic adaptation modifies separate plastic weight matrices

$$U^{\text{plastic}}$$

which allow behavioral responses to adjust through experience while preserving the inherited genetic baseline.

The effective behavioral parameters used by the agent are computed as

$$U = \frac{U^{\text{fixed}} + U^{\text{plastic}}}{2}$$

This formulation allows behavior to be shaped by both evolutionary inheritance and lifetime learning.

3.6.3 Genetic Mutation

During reproduction, each inherited parameter may undergo mutation with a small probability.

For a genetic parameter $g \in G$, mutation is applied as

$$g' = g + \epsilon$$

where

$$\epsilon \sim \mathcal{N}(0, \sigma^2).$$

The resulting value is clipped to the valid range

$$g' \in [0.05, 1.0].$$

This mutation mechanism allows the evolutionary process to explore new behavioral strategies across generations.

3.6.4 Fitness Function

Each candidate genome is evaluated over N stochastic rollouts (different random seeds) under fixed environment settings. Fitness rewards longer survival (normalized time-to-death, TTD) for child and mother, with a small penalty for child injury at episode end:

$$F = \alpha \text{TTD}_{\text{child}} + (1 - \alpha) \text{TTD}_{\text{mother}} - \lambda \text{Injury}_{\text{child}} \quad (3.2)$$

where $\text{TTD} \in [0, 1]$ is death tick divided by episode horizon, $\text{Injury}_{\text{child}}$ is mean final child injury (0–100, scaled in implementation), α is a configurable weight on child survival (default $\alpha = 0.5$ in this thesis), and $\lambda = 0.002$ discourages high-injury strategies. Evolution uses a (1+1)-style rule: propose one mutated genome, accept only if mean fitness across rollouts improves.

3.7 Simulation Scenarios and Experimental Design

This section describes how the proposed neuroendocrine-driven agents are evaluated through controlled simulation scenarios. The goal is to verify that the internal mechanisms (perception, physiological regulation, neuroendocrine dynamics, and motivational competition) produce coherent, emergent behaviors at the agent and multi-agent levels.

3.7.1 Environmental Parameters

We vary environment difficulty along two axes that are directly implemented in the code and used in experiments: food availability (food spawn interval) and predation pressure (number of threats). These parameters control the trade-off between foraging, caregiving, and protection under limited resources and risk.

Table 3.2: Environmental parameter space

Parameter	Levels	Values
Food interval	3	{5, 15, 25} ticks between food spawns
World difficulty	3	{easy, normal, hard} implemented as {0, 1, 2} threats

These settings allow us to evaluate:

- resource pressure on maternal foraging behavior,
- threat pressure on protective motivation and injury outcomes,
- robustness to distribution shift by transferring evolved genomes to unseen combinations (E4).

In this thesis, the primary evaluated setting is the dyadic mother–child scenario; multi-family interaction is discussed as a future extension for studying kinship, alloparental care, and population structure effects.

3.7.2 Experimental questions (aligned with the final presentation)

The experimental program is organized around four questions:

1. **What does “maternal behavior” look like in this world?** We operationalize maternal behavior via survival (TTD), state regulation trajectories, and need-conditional action selection such as $P(\text{Forage} \mid \text{hungry})$, $P(\text{Care} \mid \text{cold})$, and $P(\text{Protect} \mid \text{injured})$.
2. **Role of plasticity and the Baldwin effect.** We test whether lifetime plasticity improves the speed or quality of adaptation by comparing evolution-only (E2) to outcome-based plasticity regimes. In **P1**, outcome-based plastic updates are active during evolution and use an adaptive learning rate scaled by deficit size; in **P2**, the same outcome-based update rule uses a fixed learning rate and is applied at evaluation time on the evolved genome. This matches the Baldwinian hypothesis that learning can guide evolution without learned changes being directly inherited.

3. **Effect of initial maternal traits.** We compare three weight initialization regimes (anti-maternal, random-uniform, pro-maternal) and quantify how priors shape evolutionary trajectories and final behaviors under identical selection pressure.
4. **Generalization to unseen environments.** We evaluate transfer to unseen combinations of world difficulty and food availability (E4, a 3×3 grid of threat count $\times$ food interval) and summarize structured failure modes (e.g., hunger vs. cold vs. injury deaths).

Chapter 4 Experiments and Results

This chapter reports four experiments aligned with the final presentation: (1) operationalizing maternal behavior in the training ecology, (2) testing plasticity and a Baldwin-style acceleration of adaptation, (3) measuring how initialization priors shape evolutionary trajectories, and (4) evaluating generalization on a held-out world $\times$ food grid (E4 transfer).

4.1 Common experimental setup

4.1.1 Environment and episode horizon

Unless stated otherwise, each rollout uses a 15×15 grid with one mother and one child, a configurable number of threat agents, and periodic food spawning. One episode lasts up to 1000 ticks.

4.1.2 Food spawn interval (E4 grid)

The parameter **food spawn interval** is the number of ticks between food spawns (with one food item per spawn in our runs). A **shorter** interval means food appears more often: interval 5 = *abundant* food, interval 15 = *medium* (training/emergence default), interval 25 = *scarce* food. In E4 figures, the food axis is labeled 5/15/25 with 5 on the abundant side and 25 on the scarce side.

4.1.3 Evolution and fitness

Evolution follows the single-lineage (1+1) scheme described in Chapter 3. Each evolutionary replicate is one mother lineage: a single heritable motivation-weight genome evolved over 3000 generations. Each candidate genome is evaluated over $N = 10$ stochastic rollouts (different RNG seeds) and accepted only if mean fitness improves. Mutation uses $\epsilon \sim \mathcal{N}(0, 0.05)$ applied elementwise to motivation weights (clamped to $[0.05, 1.0]$).

Fitness uses Equation (3.2) with default $\alpha = 0.5$ (child-weight) and $\lambda = 0.002$ (injury penalty).

4.1.4 Initialization regimes and plasticity regimes

We compare three initialization regimes for motivation weights: `anti_maternal` (low Care/Protect, high Forage/Self), `random_uniform`, and `pro_maternal` (high Care/Protect). We also compare three adaptive regimes: **E2** (evolution-only, no within-episode weight updates), **P1** (outcome-based plasticity with an *adaptive* learning rate scaled by deficit size, global deficit signal, per-tick updates), and **P2** (outcome-based plasticity with a *fixed* learning rate, local deficit signal, per-tick updates).

4.1.5 Reported metrics

Across experiments we report: normalized child/mother TTD, mother/child state trajectories (alive-tick means), conditional action probabilities $P(\text{Forage} \mid \text{hungry})$, $P(\text{Care} \mid \text{cold})$, $P(\text{Protect} \mid \text{injured})$, genome weight summaries, and child death-cause mixtures (hunger/cold/injury/alive). Unless noted, emergence analyses use 30 independent evolution seeds; post-evolution evaluation uses 10 rollouts per final genome.

4.2 Experiment 1: What “maternal behavior” looks like

4.2.1 Goal and training condition

This experiment operationalizes “maternal behavior” as coordinated need-dependent switching between foraging, caregiving, and protection that improves child survival time (TTD) while maintaining maternal viability. The training condition uses a normal world with one threat and a medium food spawn interval (15 ticks between spawns), matching the final presentation.

4.2.2 Survival across baseline conditions

Figures 4.1 and 4.2 compare survival endpoints and full survival curves across baseline rollout conditions in that training ecology. Child TTD varies substantially across conditions: some policies keep the child alive much longer (higher median and wider upper quartile on the boxplots), while others fail early. Mother TTD shows a parallel spread, indicating that better child outcomes are not free—conditions that protect the child longer can still differ in how long the mother remains viable. Kaplan–Meier curves make the timing explicit: conditions whose child curves decay more slowly sustain survival probability over more of the episode; mother curves show whether maternal death precedes or follows child failure.

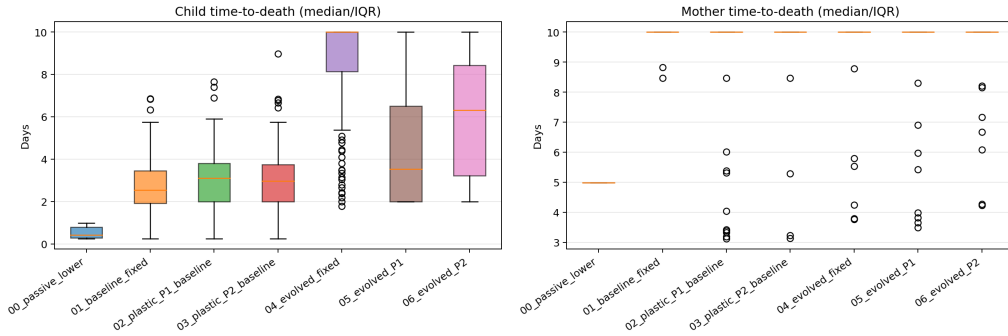


Figure 4.1: Distribution of child (left) and mother (right) time-to-death across baseline rollout conditions in the training world. Boxes show median and interquartile range (IQR) over independent episode replicates. The y-axis is survival time in days (100 simulation ticks per day).

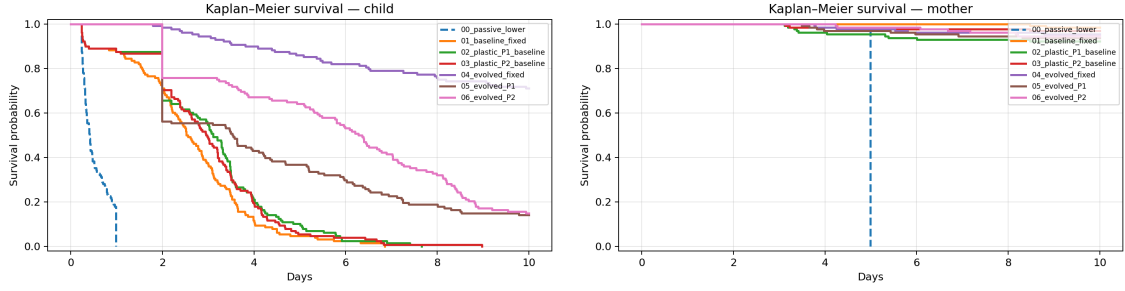


Figure 4.2: Kaplan–Meier survival curves for child (left) and mother (right) by baseline condition.

4.2.3 Episode dynamics

Figure 4.3 summarizes typical episode dynamics over 128 stochastic rollouts (mean $\pm$ std). Panel (a): mother and child can remain co-alive for a substantial fraction of the horizon when the policy is effective. Panel (b): child hunger, warmth, and injury on alive ticks—effective care keeps these in safer ranges while the child lives. Panel (c): Forage, Care, Self, and Protect are all used at non-negligible rates (motivational competition, not a frozen policy). Panel (d): mother energy, fatigue, bonding, fear_threat, stress, closeness_child, and slow endocrine traces (OT/CORT).

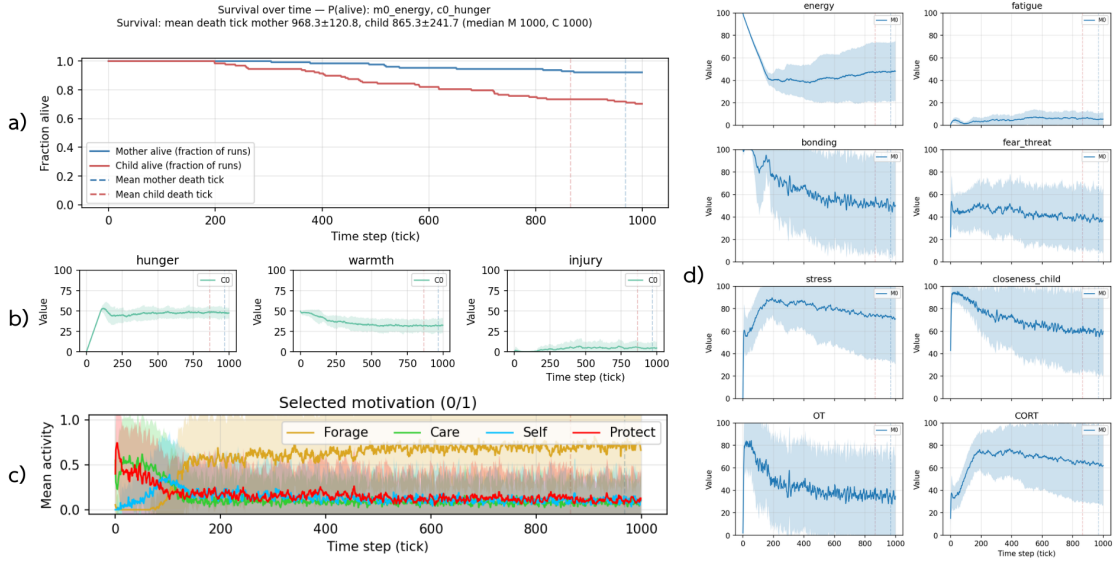


Figure 4.3: Representative rollout dynamics aggregated over 128 stochastic episode seeds (mean $\pm$ std): (a) $P(\text{alive})$; (b) child hunger, warmth, injury; (c) motivation activity; (d) mother internal states.

4.2.4 Takeaway (Experiment 1)

The training world supports non-trivial mother–child dynamics: survival outcomes separate across conditions, and successful rollouts exhibit coordinated physiology, internal

states, and motivation usage.

4.3 Experiment 2: Role of plasticity and the Baldwin effect

4.3.1 Hypothesis

Following the Baldwin effect, within-lifetime plasticity can improve behavior during an episode and can accelerate evolutionary adaptation even if learned changes are not inherited.

4.3.2 Results

Figure 4.4 sweeps world difficulty (easy, normal, hard) and child-fitness emphasis $\alpha_{\text{child}} \in \{0.3, 0.5, 0.7\}$, comparing E2, P1, and P2. In virtually every panel, plastic runs (P1 red, P2 green) rise to higher fitness earlier than E2 (blue), with the gap most visible in normal and hard worlds. The dashed traces (mean within-episode $|\Delta u_{\text{plastic}}|$) are largest in early generations and then flatten while fitness continues to improve—consistent with plasticity helping search early and inherited weights accounting for late performance. The center panel (normal world, $\alpha = 0.5$) matches the emergence setting used in Experiments 3–4.

Baldwin grid — Fitness + Mean L1 $|\Delta u_{\text{plastic}}|$ per tick (dashed, right axis) (mean $\pm$ IQR, MA window=50) [shared y: [0.2148, 0.8292]] [shared right y: [-0.001845, 0.03875]]

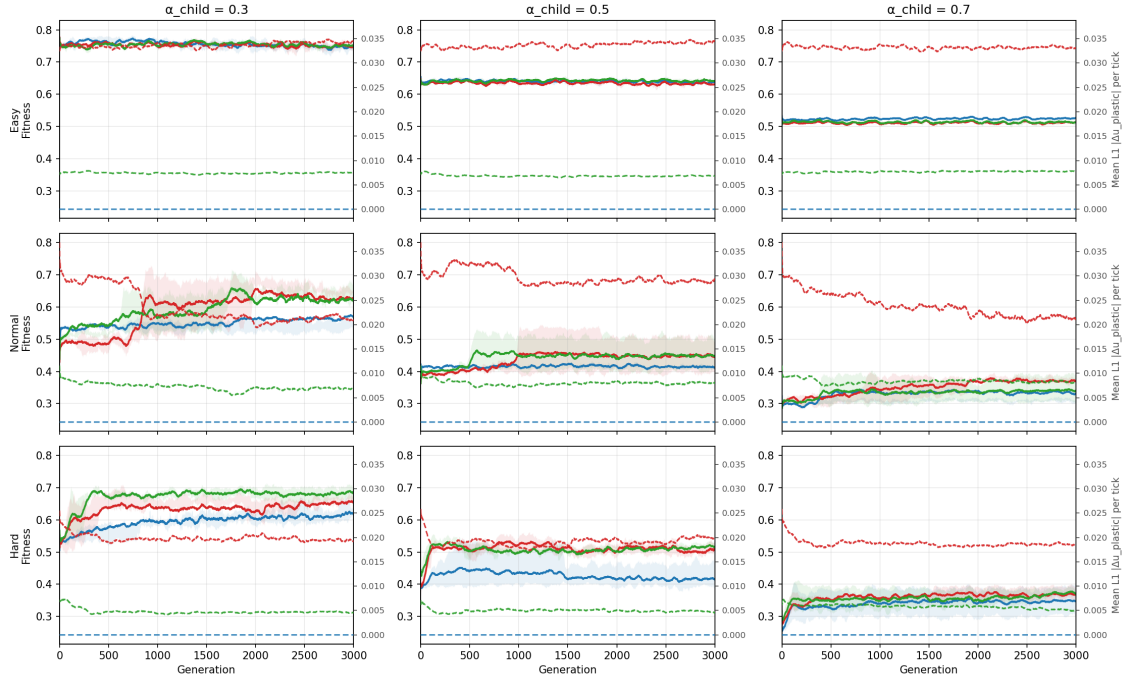


Figure 4.4: Plasticity sweep: rows = world (easy/normal/hard); columns = α_{child} (0.3/0.5/0.7). Solid lines: mean fitness (E2 blue, P1 red, P2 green; IQR bands). Dashed: mean $|\Delta u_{\text{plastic}}|$ (right axis). Center panel = emergence training setting.

4.3.3 Takeaway (Experiment 2)

Outcome-based plasticity (P1/P2) speeds evolutionary improvement relative to E2, especially under harder ecological pressure, motivating their inclusion in Experiments 3–4.

4.4 Experiment 3: Effect of initial maternal traits

4.4.1 Hypothesis

Initialization priors shape evolutionary trajectories: anti-maternal agents should shift most, random agents moderately, and pro-maternal agents least.

4.4.2 Genome drift (L_2 over 30 lineages)

Figure 4.5 plots $L_2\|u(g) - u(0)\|$ during emergence (training world; mean $\pm$ SEM over 30 seeds). Anti-maternal lineages (solid) reach the highest drift for all plasticities; pro-maternal (dotted) the least; random-uniform (dashed) in between—stable ordering anti $>$ random $>$ pro across E2, P1, and P2.

L_2 drift from gen-0 weights — mean across seeds [42_43_44_45_46_47_48_49_50_51_52_53_54_55_56_57_58_59_60_61_62_63_64_65_66_67_68_69_70_71] (MA; $\pm$ SEM)

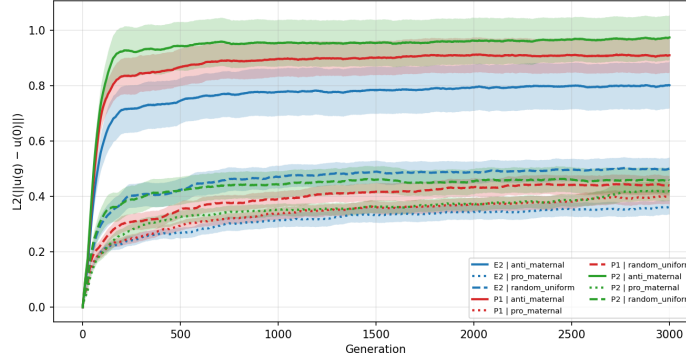


Figure 4.5: L_2 genome drift vs. generation (MA window 50; mean $\pm$ SEM over 30 lineages). Color = plasticity; linestyle = initialization. Population aggregate, not one mother.

4.4.3 Exemplar weight paths (one seed per init)

Figure 4.6 shows checkpoint heatmaps for **one** mother lineage per initialization (single (1+1) run each), not averaged over seeds. Anti-maternal runs reorganize Care/Protect leaves from low to high within the first 500–1000 generations; pro-maternal runs start near maternal profiles and change little; random runs adjust more diffusely.

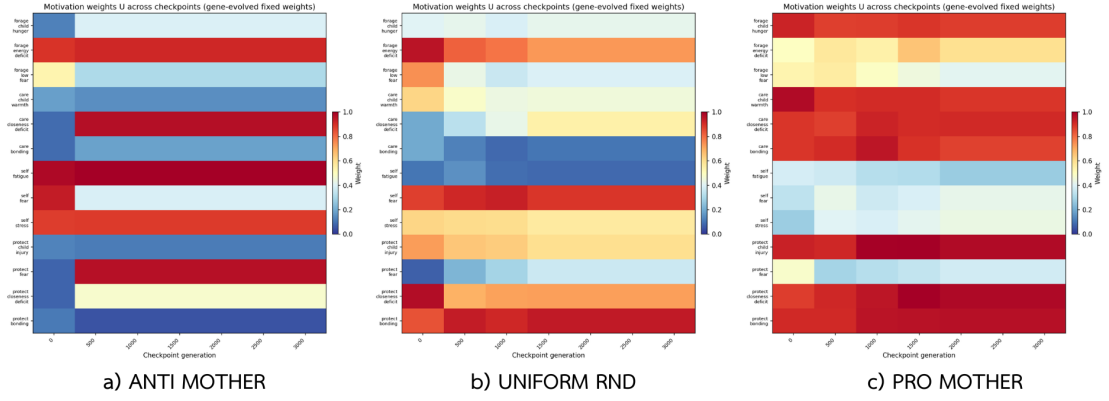


Figure 4.6: Exemplar fixed-genome weights u_{fixed} at checkpoints (0–3000, every 500 gens): one seed per init (anti / random / pro). Rows = 13 motivation leaves. Not a 30-seed average; see Figure 4.5 for population drift.

4.4.4 Per-weight and final init contrast (30 seeds)

Figure 4.7 shows selected leaves by $\text{init} \times \text{plasticity}$ cell (mean $\pm$ IQR over 30 lineages). Figure 4.8 shows final pro-minus-anti weight differences: partial convergence, with residual init signatures on several leaves.

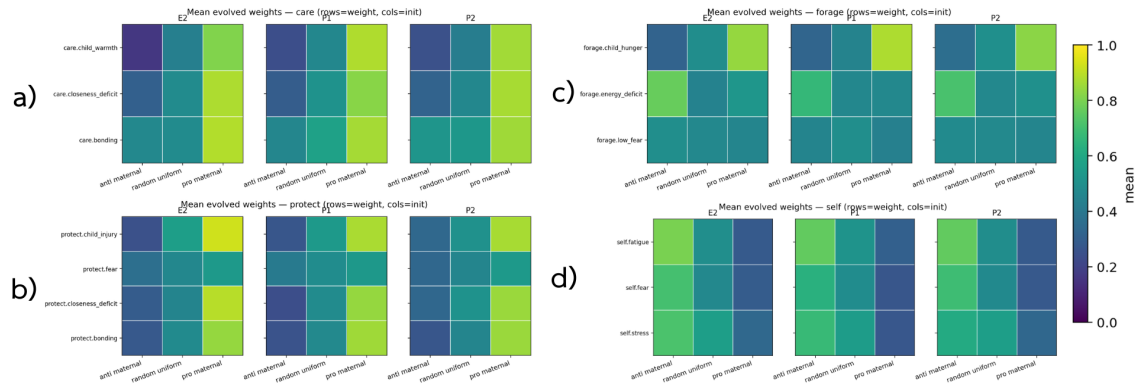


Figure 4.7: Selected motivation weights vs. generation (3 $\times$ 3: init $\times$ plasticity; mean $\pm$ IQR, 30 seeds).

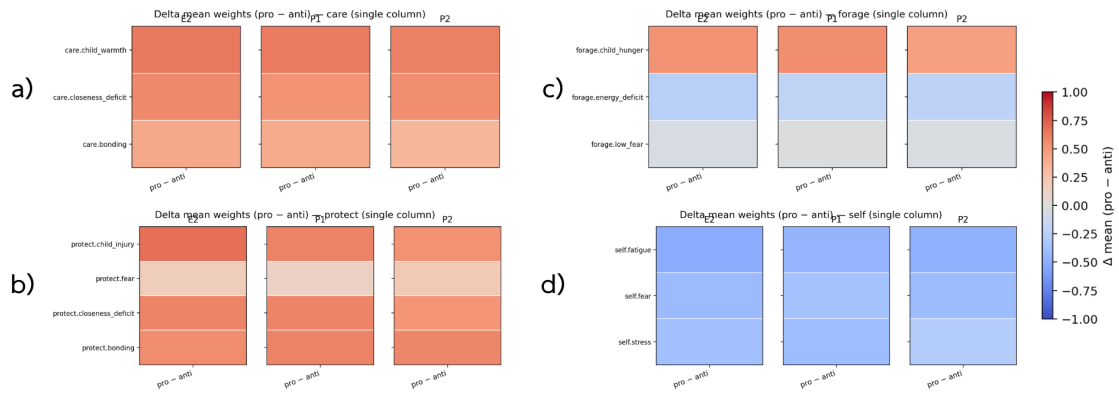


Figure 4.8: Final weights: pro_maternal minus anti_maternal (mean over 30 seeds), per plasticity panel.

4.4.5 Child survival and need-dependent behavior

Figure 4.9: normalized child TTD rises from early to emergent stages for all inits/plasticities (vs. passive and strong-active reference bands)—evolution improves child survival. Figure 4.10: conditional $P(\text{Forage} \mid \text{hungry})$, $P(\text{Care} \mid \text{cold})$, $P(\text{Protect} \mid \text{injured})$ increase toward need-appropriate switching; child physiology moves toward safer alive-tick ranges.

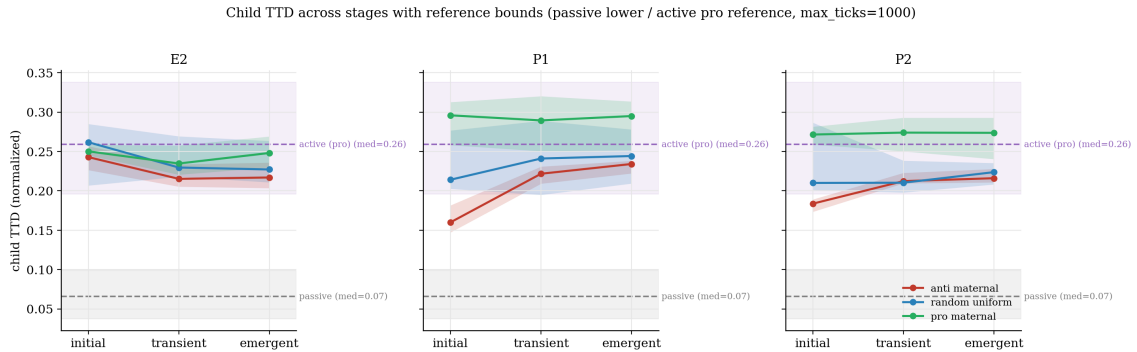


Figure 4.9: Child TTD across evolutionary stages (E2/P1/P2 panels; init-colored medians with IQR across genomes).

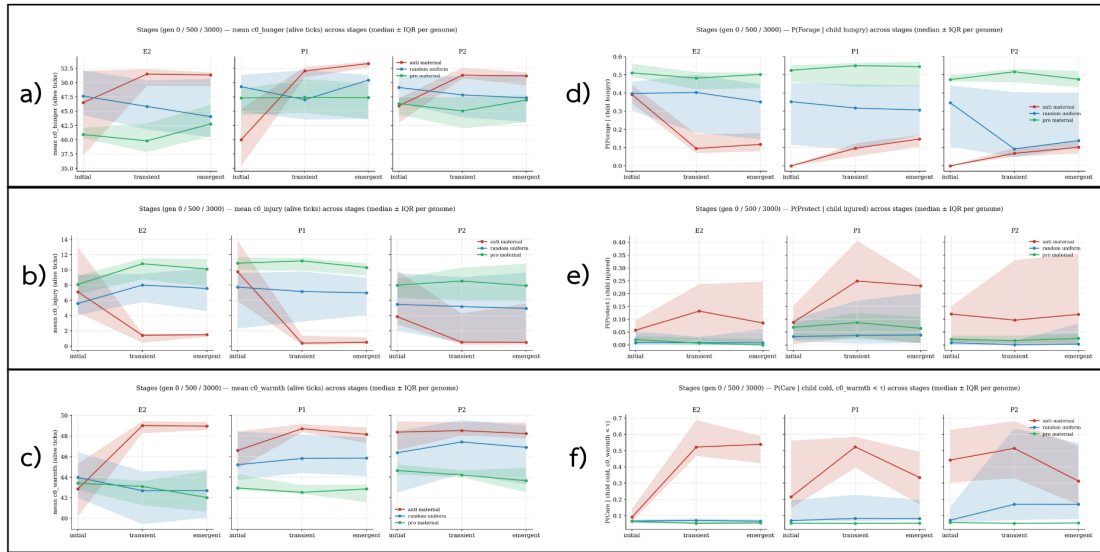


Figure 4.10: Conditional motivation probabilities and mean child hunger/warmth/injury during emergence.

4.4.6 Death causes on the seen world (normal, food interval 15)

Figures 4.11–4.12 evaluate final genomes on the *training-matched* ecology only (normal threats, medium food interval 15—not the full E4 grid). Hunger is the dominant pooled failure for anti-maternal and random-uniform genomes; pro-maternal genomes show more balanced alive/cold/injury outcomes.



Figure 4.11: Child outcomes on seen world (normal, food interval 15): per evolution seed (30 bars per panel; 10 rollouts each).

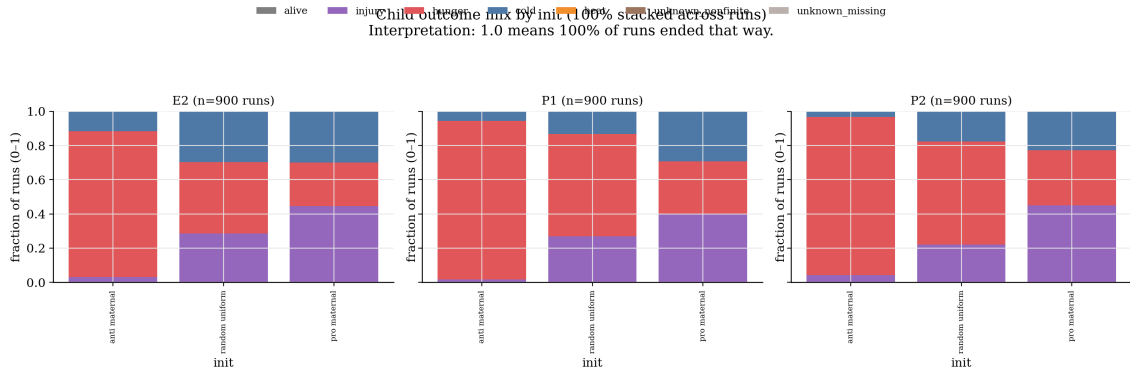


Figure 4.12: Same evaluation, pooled over seeds: three plasticity panels, one stacked bar per init.

4.4.7 Takeaway (Experiment 3)

Anti-maternal priors require the largest genome displacement; pro-maternal the smallest; all inits improve child TTD and conditional care, but hunger failures remain more common for anti-maternal genomes on the training ecology.

4.5 Experiment 4: Generalization to unseen environments (E4)

4.5.1 Unseen condition grid

Final genomes from emergence are evaluated on nine cells: world $\{easy, normal, hard\}$ $\times$ food spawn interval $\{5, 15, 25\}$, where **5 = abundant**, **15 = medium** (training), **25 = scarce**. Ten rollouts per genome per cell; 30 seeds per (plasticity, init) $\Rightarrow$ 300 episodes per tile in death-cause figures.

4.5.2 Transfer performance

Figure 4.13: performance is best on **easy** worlds and at **food 5** (abundant); worst on **hard** worlds and at **food 25** (scarce). Pro-maternal curves often sit above anti-maternal on mild cells; gaps shrink on the harshest corners.

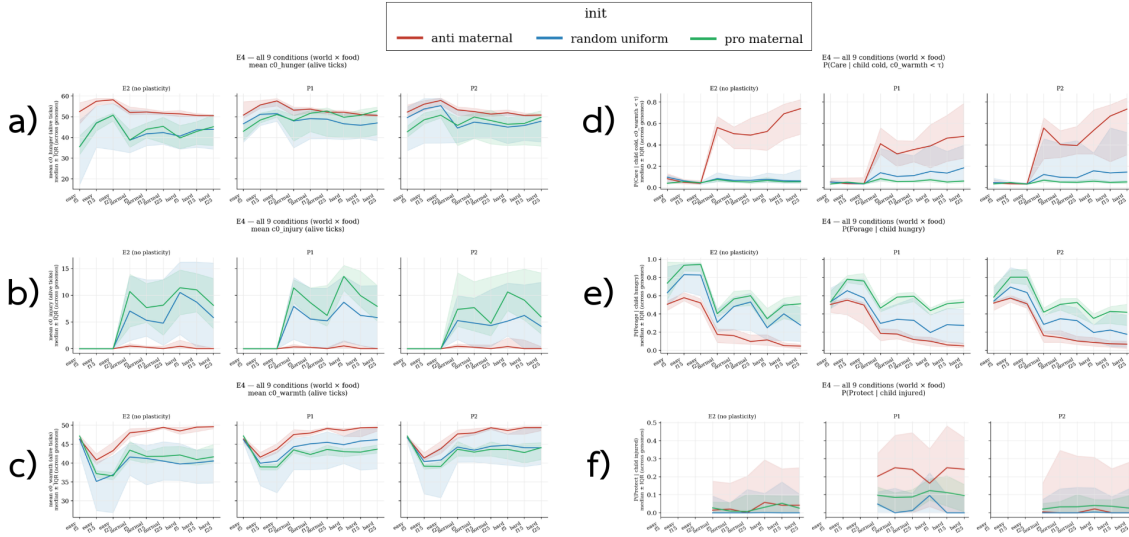


Figure 4.13: E4 transfer on the 3×3 grid (world $\times$ food interval 5/15/25; 5 = abundant, 25 = scarce). Columns: E2/P1/P2; line color: init.

4.5.3 Child and mother TTD on the grid

Figures 4.14 and 4.15: child TTD falls most on hard worlds and at food 25 (scarce); comparing panels separates child-only failure from joint mother–child mortality.

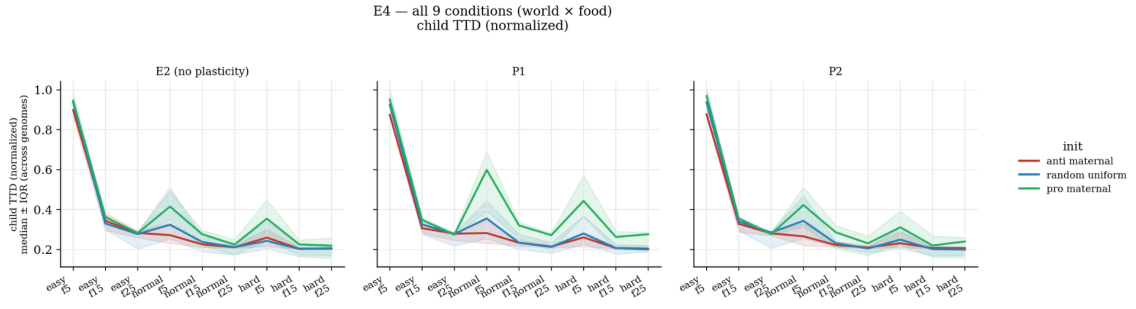


Figure 4.14: Normalized child TTD on nine E4 cells (median $\pm$ IQR over 30 genomes).

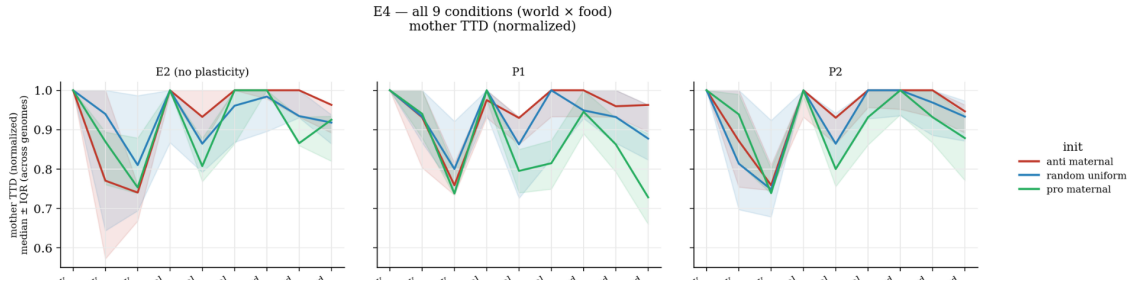


Figure 4.15: Normalized mother TTD on nine E4 cells (same layout as child panel).

4.5.4 Death causes per cell and pooled

Figure 4.16: on easy rows and at food 5, **alive** (A) is often plurality; on normal/hard rows and at food 25, **hunger** (hun) dominates many anti-maternal tiles. Figures 4.17–4.18: all nine cells pooled (90 rollouts per seed).

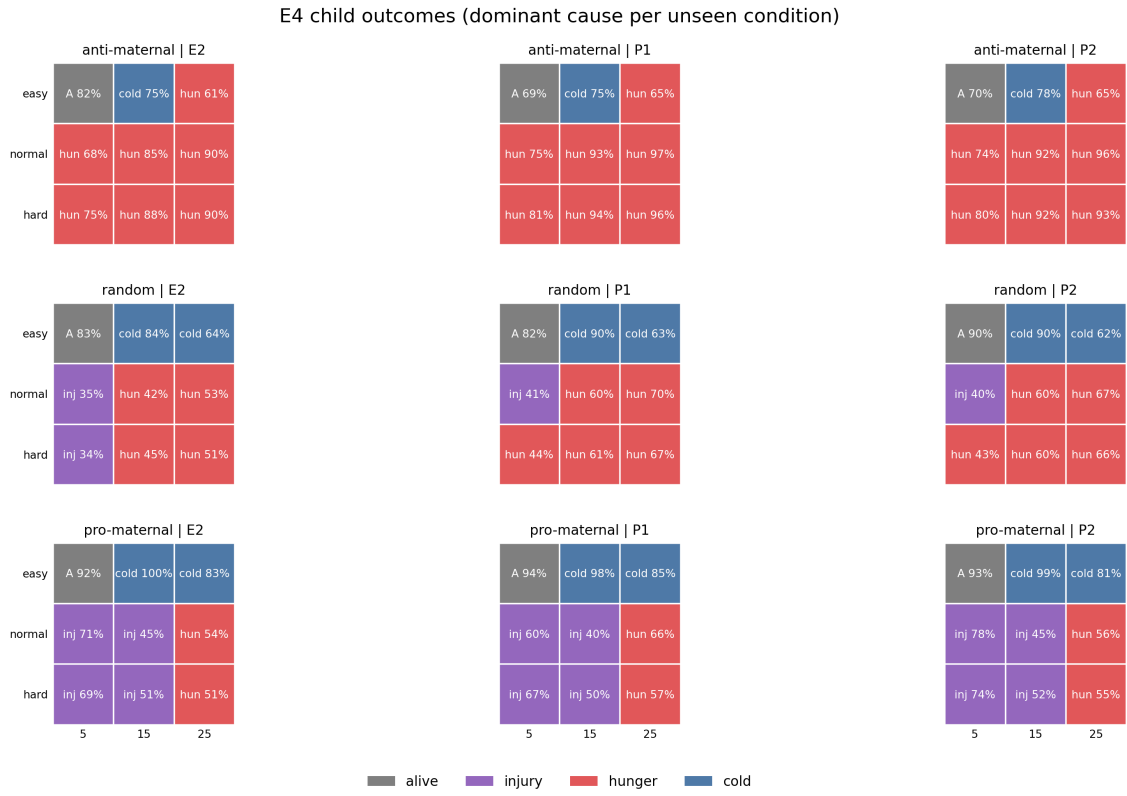


Figure 4.16: Dominant child outcome per E4 cell (A/hun/cold/inj; % over 300 rollouts). Insets: world (rows) $\times$ food 5/15/25 (cols; 5 abundant, 25 scarce).



Figure 4.17: Outcomes pooled over all nine E4 cells, per evolution seed (9×10 rollouts per bar).

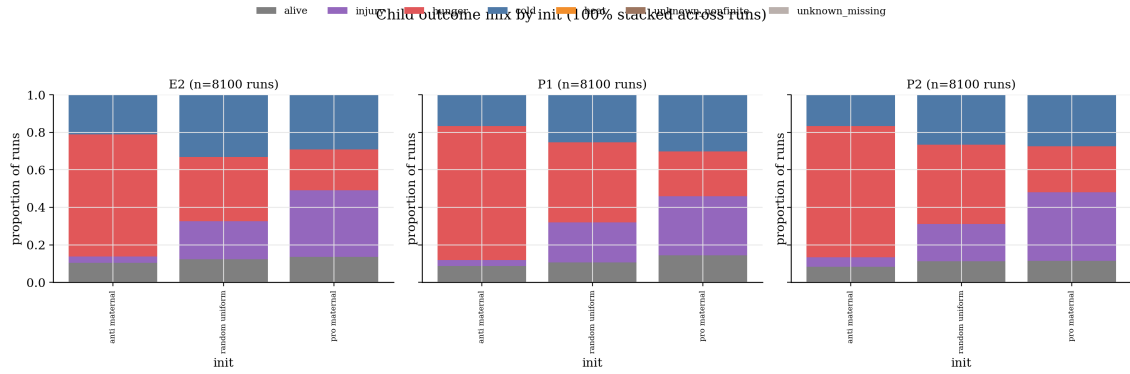


Figure 4.18: Same nine-cell pool, summarized by (plasticity, init).

4.5.5 Takeaway (Experiment 4)

Transfer is graded: easy worlds and abundant food (interval 5) preserve child survival; hard worlds and scarce food (interval 25) produce hunger-dominated failure. Agents generalize to milder transfer cells better than to the scarcest-food, highest-threat corners.

4.6 Chapter summary

Table 4.1: Summary of experimental questions and main outcomes.

Exp	Question	Main outcome
1	Maternal behavior in training world?	Survival and states differ by condition; effective policies coordinate motivations and physiology.
2	Plasticity / Baldwin?	P1/P2 outperform E2 in fitness rise; large early plastic updates.
3	Init priors shape evolution?	Anti drifts most, pro least; care/TTD improve for all; hunger bias remains for anti on seen world.
4	Generalization (E4)?	Good on easy + abundant food (5); poor on hard + scarce food (25); hunger dominates harsh cells.

Overall, need-dependent maternal policies can evolve from anti-maternal starts, plasticity speeds search, and transfer fails predictably when threat pressure and food scarcity (interval 25) exceed the training regime (normal world, interval 15).

Chapter 5 Conclusion and Discussion

This chapter summarizes the four experimental blocks reported in Chapter 4 (Experiment 1–4), relates them to the research questions from Chapter 1, and discusses limitations and future work. Food spawn intervals are stated consistently throughout: interval 5 = *abundant* food, interval 15 = *medium* (training/emergence), interval 25 = *scarce* food.

5.1 Summary of findings by experiment

5.1.1 Experiment 1: Maternal behavior in the training ecology

Experiment 1 asked what “maternal behavior” looks like in the normal training world (one threat, food spawn interval 15) before evolution. Baseline rollouts showed separated child and mother survival distributions (boxplots) and distinct Kaplan–Meier decay rates across conditions. Aggregated episode traces (128 seeds) showed co-survival periods, child hunger/warmth/injury on alive ticks, concurrent use of Forage/Care/Self/Protect, and coupled mother internal states (energy, fatigue, bonding, fear, stress, closeness, OT/CORT). **Conclusion:** the environment and motivational architecture support non-degenerate, state-dependent mother–child dynamics; later experiments can therefore interpret fitness gains as changes in real behavioral regimes rather than artifacts of a trivial simulator.

5.1.2 Experiment 2: Plasticity and the Baldwin effect

Experiment 2 swept world difficulty (easy/normal/hard) and child-fitness weight $\alpha_{\text{child}} \in \{0.3, 0.5, 0.7\}$ for E2 (genes only), P1 (outcome-based plasticity, adaptive learning rate, global deficits), and P2 (outcome-based plasticity, fixed learning rate, local deficits). Across panels, P1 and P2 reached higher mean fitness earlier than E2, with the largest gaps in normal and hard worlds; within-episode plastic weight updates were strongest in early generations and then flattened while fitness continued to improve. The normal-world, $\alpha = 0.5$ panel matches the emergence runs in Experiments 3–4. **Conclusion:** lifetime plasticity acts as a *search prior* consistent with a Baldwin-style account—plasticity helps early, inherited motivation weights carry late performance—motivating inclusion of P1/P2 alongside E2 when testing initialization and transfer.

5.1.3 Experiment 3: Initial maternal traits and emergence on the seen world

Experiment 3 tested whether initialization priors (*anti_maternal*, *random_uniform*, *pro_maternal*) shape evolutionary paths when selection pressure is held fixed (30 seeds, 3000 generations, training world).

Genome change. Population L_2 drift from generation-0 weights (mean $\pm$ SEM over 30 lineages) was ordered anti $>$ random $>$ pro for all plasticities—priors set how far selection must move the genome. Exemplar checkpoint heatmaps (one seed per init) illustrated the mechanism: anti lineages reorganize Care/Protect leaves from low to high; pro lineages start near maternal profiles and change little. Final pro-minus-anti weight maps showed *partial* convergence (several leaves still differ) rather than identical endpoints.

Outcomes and behavior. Child normalized TTD improved from early to emergent stages for all inits/plasticities (relative to passive and strong-active reference bands). Conditional probabilities $P(\text{Forage} \mid \text{hungry})$, $P(\text{Care} \mid \text{cold})$, and $P(\text{Protect} \mid \text{injured})$ rose toward need-appropriate switching—maternal-like policies can be *selected into* anti-maternal genomes.

Failure modes on the training-matched evaluation (normal, food interval 15). Post-evolution rollouts on the *seen* ecology (not the full E4 grid) showed that hunger remained the dominant child failure for anti-maternal and random-uniform genomes, while pro-maternal genomes had more balanced alive/cold/injury outcomes. Initialization therefore affects both *how* evolution proceeds and *which* residual failure channel remains after 3000 generations.

Conclusion: H1 is supported (anti shifts most, pro least); partial behavioral convergence occurs, but init signatures persist in weights and death-cause mixtures on the training world.

5.1.4 Experiment 4: Generalization (E4 transfer grid)

Experiment 4 evaluated final genomes on nine held-out cells: world $\times$ food interval $\{5, 15, 25\}$ with 10 rollouts per genome per cell.

Transfer was **graded**, not all-or-none. Easy worlds and abundant food (interval 5) yielded high child survival and frequent “alive” plurality outcomes in the per-cell death-cause tiles. Hard worlds and scarce food (interval 25) produced sharp drops in child TTD and hunger-dominated tiles, especially for anti-maternal genomes. Normal-world cells at medium food (interval 15, training-matched) still showed substantial hunger for anti-maternal inits—performance on the training cell does not imply robustness to scarcer food or harder threats. Pooled nine-cell death summaries integrated $9 \times 10 = 90$ rollouts per seed and confirmed the same init/plasticity ordering at a global level: more alive mass for pro-maternal, more hunger for anti-maternal.

Conclusion: evolved policies adapt to the *niche* shaped by emergence (normal threats, interval 15); they generalize to easier transfer corners but fail predictably when threat pressure and food scarcity (interval 25) exceed training.

5.2 Answers to the research questions

Table 5.1 maps the supporting questions from Chapter 1 to the experimental evidence.

Table 5.1: Research questions and corresponding findings (Chapters 4–5).

Question theme	Finding
Internal states and hormones modulate behavior?	Exp 1 traces: OT/CORT, bonding, fear co-vary with motivations and survival; Exp 3 conditional $P(\text{sel} \mid \text{need})$ rises under selection.
Food and threats reshape Forage/Care/-Self/Protect?	Exp 4: abundant food (5) vs. scarce (25) and easy vs. hard worlds shift survival and death causes; hunger dominates harsh/scarce cells.
Generalization to unseen conditions?	Partial (Exp 4): good on easy+abundant; poor on hard+scarce; training cell $\neq$ full grid.
Anti/random/pro improve under selection?	Yes (Exp 3): TTD and conditional care improve; anti requires largest weight drift.
E2 vs. P1 vs. P2?	Exp 2: P1/P2 faster fitness rise; Exp 3–4: plasticity modulates outcomes but init ordering often persists.
Minimal computational abstraction?	(1+1) evolution of 13 motivation weights + internal states + argmax motivations suffices for emergent care-like switching (with limits below).

The **central question**—whether prosocial maternal behavior can arise without hand-crafted cooperative rewards—is answered *qualified yes* within this dyadic grid world: behavior is not programmed as “always care,” but emerges from motivational competition, internal regulation, and selection on child/mother survival. Outside the training niche, that emergence is incomplete.

5.3 Mechanistic interpretation

Three mechanisms jointly explain the figure patterns.

Motivational competition. At each tick the mother selects the highest activated motivation among Forage, Care, Self, and Protect. Need-appropriate switching appears in Exp 3 as rising $P(\text{Care} \mid \text{cold})$ and $P(\text{Protect} \mid \text{injured})$ rather than a single frozen action.

Internal regulation. Energy, fatigue, bonding, fear, stress, and hormone-like signals (OT/CORT) provide context for those activations (Exp 1). They offer an interpretable account of *why* weight changes alter behavior, not only that fitness increases.

Evolution and plasticity. Inherited weights evolve by (1+1) search: Gaussian mutation, $N=10$ rollout evaluation, accept if mean fitness improves. Plasticity (P1/P2) changes the *rate* of fitness improvement (Exp 2); initialization changes the *path length* in weight space (Exp 3). Neither removes environmental limits on transfer (Exp 4).

5.4 Limitations

- **Expressivity:** evolution tunes 13 scalar motivation weights, not policies for navigation or threat avoidance; the child is passive (no signaling genome).
- **Single-lineage search:** no population-based GA; results describe hill-climbing with stochastic evaluation noise, not global optima.
- **Fixed training niche:** emergence uses one world (normal, interval 15); E4 tests transfer but does not co-evolve environments.
- **Fitness weighting:** α_{child} and λ fix the child–mother–injury trade-off; the Exp 2 sweep shows conclusions depend on α .
- **Stochastic evaluation:** 10 rollouts per candidate and per post-hoc eval; death-cause tiles pool 300 episodes per cell—rare outcomes may be under-sampled.
- **Exemplar heatmaps:** Figure 4.6 in Chapter 4 shows one seed per init; population claims rely on the 30-seed aggregates.

5.5 Future directions

- **Statistical rigor:** bootstrap CIs on median TTD, per-tile tests on death-cause rates, and explicit comparison of P1 vs. P2 beyond visual curves.
- **Replicate exemplars:** checkpoint heatmaps for all 30 seeds or a small stratified sample to separate init effects from seed noise.
- **Richer ecology:** variable grid size, mobile threats, partial observability, and child signaling genomes.
- **Population evolution:** multi-lineage or spatially structured groups to study kin selection and alloparental care.
- **Curriculum training:** train across the E4 grid (or a curriculum from abundant to scarce food) and re-test whether hunger collapse on interval 25 diminishes.
- **Seen-world baselines:** systematic collection of final-policy rollouts on the training ecology (not only transfer-grid evaluations) to separate training-matched evaluation from out-of-distribution transfer in one pipeline.

5.6 Closing remarks

Taken together, the four experiments show that maternal-like behavior—need-conditional motivations, improved child survival under selection, and structured failure modes in harsh ecologies—can emerge from a compact biologically inspired architecture without a hand-written “care” reward. Plasticity accelerates this process; initialization shapes its trajectory; and generalization remains the primary bottleneck when food is scarce (interval 25) and threats are high. For the broader motivation of the thesis (intrinsic prosocial motivation in artificial agents), the results support *evolutionary emergence in a controlled niche*

rather than *universal robust care*. Extending environment diversity, evaluation rigor, and representational richness are the natural next steps toward claims that would apply outside the laboratory grid.

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